

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	B	B	B	A	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C	B,C,D	A,B	B,C	A,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	25.60	3.65 to 3.85	3.13 to 3.15	1.77 to 1.79	0.62 to 0.64	6.40

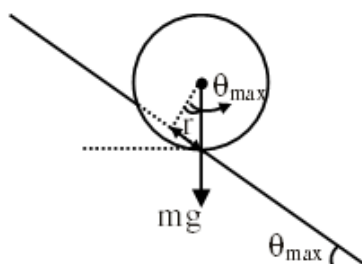
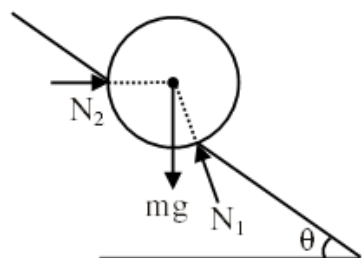
PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	B	A	C	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B,C	B,D	A,C	A,B,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.11	0.25	13.00 to 13.60	6.00 to 6.20	1.19 to 1.21	18.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	C	A	A	B	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C	B,C,D	B,C	B,C	A,B	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	1.00	1.00	108.00	5.00	1.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (A)**


For $\theta_{\max}$, the football is about to roll, then $N_2 = 0$ and all the forces (Mg and N_1) must pass through contact point

$$\therefore \cos(90^\circ - \theta_{\max}) = \frac{r}{R} \Rightarrow \sin \theta_{\max} = \frac{r}{R}$$

 2. **Ans (B)**

When the magnet is moved, it creates a state where the plate moves through the magnetic flux, due to which an electromotive force is generated in the plate and eddy currents are induced. These currents are such that it opposes the relative motion $\Rightarrow$ disc will rotate in the direction of rotation of magnet.

Note : This apparatus is called Arago's disk.

3. **Ans (B)**

For no slipping at the ground,

$$V_{\text{centre}} = \omega R \text{ (R is radius of roller)}$$

$\therefore$ Velocity of scale = $(V_{\text{center}} + \omega r)$ [r is radius of axle]

$$\text{Given, } V_{\text{center}} \cdot t = 50 \text{ cm}$$

$$\therefore \text{Distance moved by scale} = (V_{\text{center}} + \omega r)t$$

$$= \left(V_{\text{center}} + \frac{V_{\text{center}} r}{R} \right) t = \frac{3 V_{\text{center}}}{2} \cdot t = 75 \text{ cm}$$

Therefore relative displacement (with respect to centre of roller) is $(75 - 50) \text{ cm} = 25 \text{ cm}$.

4. **Ans (B)**

Torque experienced by circular loop = $\vec{M} \times \vec{B}$

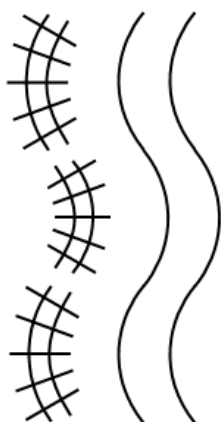
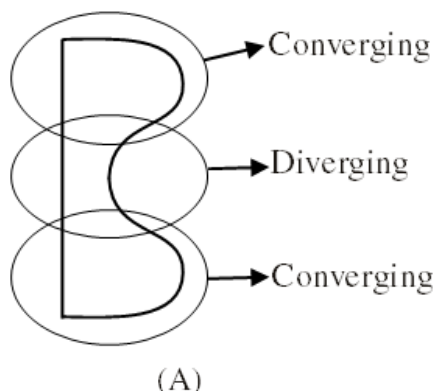
where $\vec{M}$ is magnetic moment

$\vec{B}$ is magnetic field

$$\therefore \tau = i\pi R^2 N B_0 \text{ [at the instant shown } \theta = \pi/2]$$

$$\therefore \vec{\tau} dt = d\vec{L} = i\pi R^2 N B_0 dt = Q\pi R^2 N B_0 \text{ [} i dt = Q]$$

5. **Ans (A)**



6. **Ans (B)**

$$h_1 + h_2 = 0.29 \times 2 + 0.1$$

$$h_1 + h_2 = 0.68 \text{(1)}$$

$$\Rightarrow P_0 + \rho_k g(0.1) + \rho_w g(h_1 - 0.1) \text{ [}\rho_k = \text{density}$$

of kerosene & ρ_w = density of water]

$$-\rho_w g h_2 = P_0$$

$$\Rightarrow \rho_k g(0.1) + \rho_w g h_1 - \rho_w g \times (0.1)$$

$$= \rho_w g h_2$$

$$\Rightarrow 800 \times 10 \times 0.1 + 1000 \times 10 \times h_1$$

$$- 1000 \times 10 \times 0.1 = 1000 \times 10 \times h_2$$

$$\Rightarrow 10000 (h_1 - h_2) = 200$$

$$\Rightarrow h_1 - h_2 = 0.02 \text{(2)}$$

$$\Rightarrow \boxed{h_1 = 0.35}$$

$$\Rightarrow \boxed{h_2 = 0.33}$$

$$\text{So, } \boxed{\frac{h_1}{h_2} = \frac{35}{33}}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (B,C)**

$$U = Fr$$

[Using U = Potential energy and v = velocity,

to avoid confusion between their symbols]

$$\Rightarrow \text{Force} = \frac{-dU}{dr} = -F$$

$$\Rightarrow \text{Magnitude of force} = \text{Constant} = F$$

$$\Rightarrow F = \frac{mv^2}{R} \quad \dots\dots(1)$$

$$\Rightarrow mvR = \frac{nh}{2\pi} \quad \dots\dots(2)$$

$$\Rightarrow F = \frac{m}{R} \times \frac{n^2 h^2}{4\pi^2} \times \frac{1}{m^2 R^2}$$

$$\Rightarrow R = \left(\frac{n^2 h^2}{4\pi^2 mF} \right)^{1/3} \quad \dots\dots(3)$$

$$\Rightarrow v = \frac{nh}{2\pi mR}$$

$$\Rightarrow v = \frac{nh}{2\pi m} \left(\frac{4\pi^2 mF}{n^2 h^2} \right)^{1/3}$$

$$\Rightarrow v = \frac{n^{1/3} h^{1/3} F^{1/3}}{2^{1/3} \pi^{1/3} m^{2/3}} \quad \dots\dots(4)$$

(B) is correct

$$\Rightarrow E = \frac{1}{2}mv^2 + U$$

$$= \frac{1}{2}mv^2 + FR$$

$$\Rightarrow E = \frac{1}{2}m \left(\frac{n^{2/3} h^{2/3} F^{2/3}}{2^{2/3} \pi^{2/3} m^{4/3}} \right) + F \times \left(\frac{n^2 h^2}{4\pi^2 mF} \right)^{1/3}$$

$$\Rightarrow E = \left(\frac{n^2 h^2 F^2}{4\pi^2 m} \right)^{1/3} \left[\frac{1}{2} + 1 \right]$$

$$= \frac{3}{2} \left(\frac{n^2 h^2 F^2}{4\pi^2 m} \right)^{1/3}$$

8. **Ans (B,C,D)**

$A = 64 \text{ mm}^2$, $T = 2500 \text{ K}$ (A = surface area of filament, T = temperature of filament, d is distance of bulb from observer, R_e = radius of pupil of eye)

Point source $d = 100 \text{ m}$

$$R_e = 3 \text{ mm}$$

$$(A) P = \sigma A e T^4$$

$$= 5.67 \times 10^{-8} \times 64 \times 10^{-6} \times 1 \times (2500)^4$$

($e = 1$ black body)

$$= 141.75 \text{ W}$$

Option (A) is wrong

(B) Power reaching to the eye

$$= \frac{P}{4\pi d^2} \times (\pi R_e^2)$$

$$= \frac{141.75}{4\pi \times (100)^2} \times \pi \times (3 \times 10^{-3})^2$$

$$= 3.189375 \times 10^{-8} \text{ W}$$

Option (B) is correct

$$(C) \lambda_m T = b$$

$$\lambda_m \times 2500 = 2.9 \times 10^{-3}$$

$$\Rightarrow \lambda_m = 1.16 \times 10^{-6}$$

$$= 1160 \text{ nm}$$

Option (C) is correct

(D) Power received by one eye of observer

$$= \left(\frac{hc}{\lambda} \right) \times \dot{N}$$

$\dot{N}$ = Number of photons entering into eye per second

$$\Rightarrow 3.189375 \times 10^{-8}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1740 \times 10^{-9}} \times \dot{N}$$

$$\Rightarrow \dot{N} = 2.79 \times 10^{11}$$

Option (D) is correct

9. Ans (A,B)

Given $L = x^\alpha$ (1)

$LT^{-1} = x^\beta$ (2)

$LT^{-2} = x^p$ (3)

$MLT^{-1} = x^q$ (4)

$MLT^{-2} = x^r$ (5)

$\frac{(1)}{(2)} \Rightarrow T = x^{\alpha-\beta}$

From (3)

$\frac{x^\alpha}{x^{2(\alpha-\beta)}} = x^p$

$\Rightarrow \alpha + p = 2\beta$ (A)

From (4)

$M = x^{q-\beta}$

From (5) $\Rightarrow x^q = x^r x^{\alpha-\beta}$

$\Rightarrow \alpha + r - q = \beta$ (6)

Replacing value ' α ' in equation (6) from (A)

$2\beta - p + r - q = \beta$

$\Rightarrow p + q - r = \beta$ (B)

Replacing value of ' β ' in equation (6) from (A)

$2\alpha + 2r - 2q = \alpha + p$

$\alpha = p + 2q - 2r$

10. Ans (B,C)

$a_y = -400\sqrt{3} \times 10^{10} [qE_y = ma_y]$

$R = 5 = \frac{40 \times 10^{12} \sin 2\theta}{400\sqrt{3} \times 10^{10}} \left[R(\text{range}) = \frac{u^2 \sin 2\theta}{a_y} \right]$

$\sin 2\theta = \frac{\sqrt{3}}{2}$

$2\theta = 60^\circ, 120^\circ \Rightarrow \theta = 30^\circ, 60^\circ$

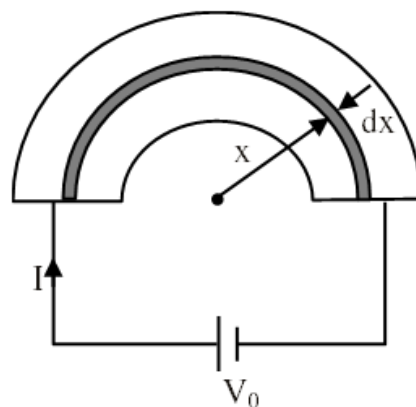
Time of flight $T_1 = \frac{2 \times 2\sqrt{10} \times 10^6 \times \frac{1}{2}}{400\sqrt{3} \times 10^{10}} = \sqrt{\frac{5}{6}} \mu s$

(for $\theta = 30^\circ$)

Time of flight $T_2 = \frac{2 \times 2\sqrt{10} \times 10^6 \times \frac{\sqrt{3}}{2}}{400\sqrt{3} \times 10^{10}} = \sqrt{\frac{5}{2}} \mu s$

(for $\theta = 60^\circ$)

11. Ans (A,C,D)



All the elements are in parallel

$\therefore \int \frac{1}{dr} = \int_{R_1}^{R_2} \frac{t dx}{\rho \pi x}$

$\frac{1}{r} = \frac{t}{\pi \rho} \ln \left(\frac{R_2}{R_1} \right)$

Resistance = $\frac{\pi \rho}{t \ln \left(\frac{R_2}{R_1} \right)}$

$i = \frac{V_0 t \ln \left(\frac{R_2}{R_1} \right)}{\pi \rho}$ (A)

$(-e\vec{E})$ will be inward direction in order to provide centripetal acceleration. Therefore electric field will be radially outward

$V_{\text{outer}} < V_{\text{inner}}$ (C)

$\frac{mV_d^2}{r} = qE$

$E = \frac{mV_d^2}{qr}$ ($I = neAV_d \Rightarrow V_d \propto i$)

$\Delta V = \int \vec{E} \cdot d\vec{r}$

$\Delta V \propto V_d^2$

$\Delta V \propto I^2$

12. Ans (A,B,C)

$$n_1 \gg (n_1 - n_2) = \Delta n$$

$$p_1 = \frac{n_1 RT}{N_A} \quad p_2 = \frac{n_2 RT}{N_A}$$

$$F = (n_1 - n_2) k_B T S = \Delta n k_B T S \quad (A)$$

$$V = \frac{\Delta n k_B T S}{\beta}$$

Force balance $\Rightarrow$ Pressure $\times$ Area = Total number of molecules $\times \beta v$

$$\Delta n k_B T S = \ell n_1 S \beta v$$

$$\Rightarrow n_1 \beta v \ell = \Delta n k_B T \quad (B)$$

$$\text{Total number of molecules/sec} = \frac{(n_1 v dt) S}{dt}$$

$$= n_1 v S = \frac{\Delta n k_B T v S}{\beta v \ell}$$

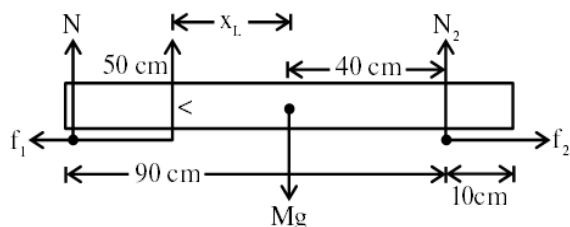
$$= \left(\frac{\Delta n}{\ell} \right) \left(\frac{k_B T}{\beta} \right) S \quad (C)$$

As Δn will decrease with time therefore rate of molecules getting transfer decreases with time.

PART-1 : PHYSICS

SECTION-II

1. Ans (25.60)



Initially

$$N_1 + N_2 = Mg \quad \left| \quad N_1 = \frac{4Mg}{9} \right.$$

$$(\tau_N = 0)_{\text{about centre}} N_1 (50) = N_2 (40) \quad \left| \quad N_2 = \frac{5Mg}{9} \right.$$

$$5N_1 = 4N_2$$

$$f_{1K} = \mu_K N_1$$

$$f_{1L} = \mu_s N_1$$

$$f_{1K} = 0.32 N_1$$

$$f_{1L} = 0.4 N_1$$

$$f_{2K} = 0.32 N_2$$

$$f_{2L} = 0.4 N_2$$

Suppose x_L = distance of left finger from centre when right finger starts moving

$$(\tau_n = 0)_{\text{about centre}} \Rightarrow N_1 x_L = N_2 (40)$$

$$f_{K1} = f_{L2} \Rightarrow 0.32 N_1 = 0.40 N_2$$

$$4N_1 = 5N_2$$

$$N_1 x_L = \frac{4N_1}{5} (40)$$

$$x_L = 32$$

Now x_R = distance when right finger stops and left finger starts moving

$$(\tau_n = 0)_{\text{about centre}} \Rightarrow N_1 x_L = N_2 (x_R)$$

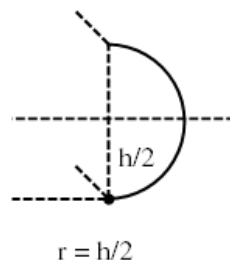
$$f_{L1} = f_{K2} \Rightarrow 0.4 N_1 = 0.32 N_2$$

$$5N_1 = 4N_2$$

$$\frac{4N_2}{5} (32) = N_2 x_R$$

$$x_R = \frac{128}{5} = 25.6 \text{ cm}$$

2. Ans (3.65 to 3.85)



Pressure at the bottom of disc = pressure due to surface tension

$$\rho gh = T \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$R_1 \gg R_2$$

$$\text{So } \frac{1}{R_1} \ll \frac{1}{R_2} \text{ and } R_2 = h/2$$

$$\therefore \rho gh = T \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = T \left[0 + \frac{1}{h/2} \right]$$

$$h^2 = \frac{2T}{\rho g}$$

$$h = \sqrt{\frac{2T}{\rho g}} = \sqrt{\frac{2 \times 0.07}{10^3 \times 10}} = \sqrt{\frac{14 \times 100}{10^4 \times 100}}$$

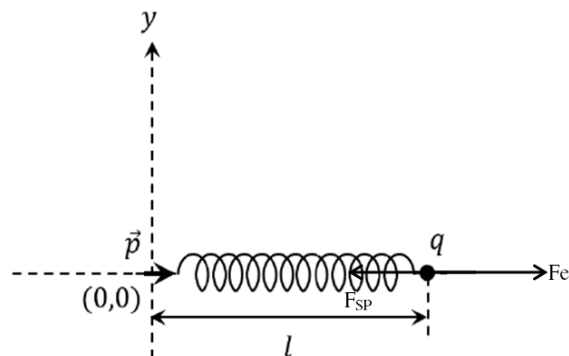
$$h = \sqrt{14} \text{ mm} = 3.741$$

3. Ans (3.13 to 3.15)

$$\Delta \ell \rightarrow x$$

$$\text{At } \ell : F_e = F_{sp}$$

$$k\ell = \frac{2kpq}{\ell^3}$$



$$F_{net} = F_{sp} - F_e = k(\ell + x) - \frac{q(2kp)}{(\ell + x)^3}$$

$$= k(x + \ell) - \frac{q(2kp)}{\ell^3 (1 + x/\ell)^3}$$

$$= kx + k\ell - q \left(\frac{2kp}{\ell^3} \right) \left(1 - \frac{3x}{\ell} \right)$$

$$= kx + k\ell - q \left(\frac{2kp}{\ell^3} \right) + \frac{2kpq}{\ell^3} \cdot \frac{3x}{\ell}$$

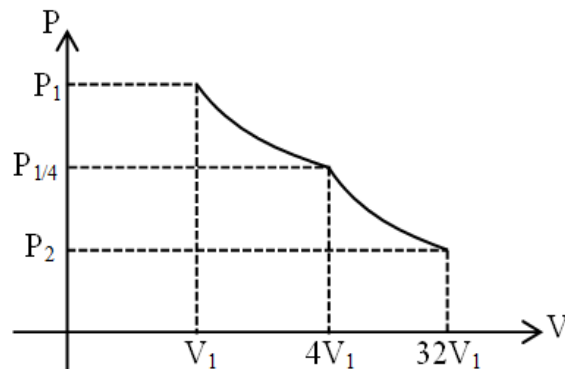
$$F_N = kx + k\ell \left(\frac{3x}{\ell} \right) = 4kx$$

$$k_{eq} = 4k \quad T = 2\pi \sqrt{\frac{m}{4k}} = \pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{\pi} \sqrt{\frac{k}{m}}$$

$$\text{So } \delta = \pi = 3.14$$

4. Ans (1.77 to 1.79)



$$\frac{P_1}{4} (4V_1)^{5/3} = P_2 (32V_1)^{5/3}$$

$$P_2 = \frac{P_1}{4} \left(\frac{1}{8} \right)^{5/3} = \frac{P_1}{128}$$

$$W_{adi} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{P_1 V_1 - \frac{P_1}{128} (32V_1)}{\frac{5}{3} - 1}$$

$$= \frac{P_1 V_1 (3/4)}{2/3} = \frac{9}{8} P_1 V_1$$

$$W_{iso} = P_1 V_1 \ln \left(\frac{4V_1}{V_1} \right) = 2P_1 V_1 \ln 2$$

$$\frac{W_{iso}}{W_{adio}} = \frac{2P_1 V_1 \ln 2}{\frac{9}{8} P_1 V_1} = \frac{16}{9} \ln 2 = f \ln 2$$

$$f = \frac{16}{9} = 1.7778 \approx 1.78$$

5. Ans (0.62 to 0.64)

$$f \propto \frac{1}{\ell_1} \Rightarrow f = \frac{k}{\ell_1} \quad \dots (1)$$

($\ell_1 \Rightarrow$ initial length of pipe)

$$\left(\frac{V}{V - V_T} \right) f = \frac{k}{\ell_2}$$

{ V_T Speed of tuning fork, $\ell_2 \rightarrow$ new length of pipe}

$$\dots (2)$$

$$(1) \div (2)$$

$$\frac{V - V_T}{V} = \frac{\ell_2}{\ell_1}$$

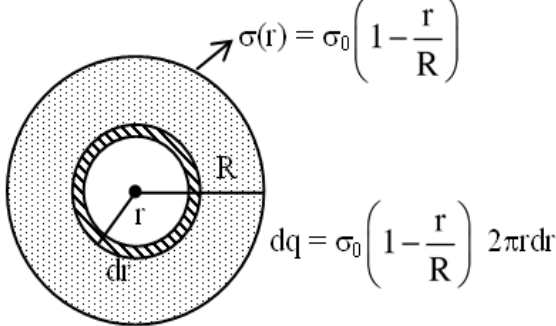
$$\frac{\ell_2}{\ell_1} - 1 = \frac{V - V_T}{V} - 1$$

$$\frac{\ell_2 - \ell_1}{\ell_1} = \frac{-V_T}{V}$$

$$\frac{\ell_2 - \ell_1}{\ell_1} \times 100 = \frac{-2}{320} \times 100 = -0.625$$

Therefore smallest value of percentage change required in the length of pipe is 0.625

6. Ans (6.40)



$$\sigma(r) = \sigma_0 \left(1 - \frac{r}{R}\right)$$

$$dq = \sigma_0 \left(1 - \frac{r}{R}\right) 2\pi r dr$$

$$\phi_0 = \frac{\int dq}{\epsilon_0} = \frac{\int_0^R \sigma_0 \left(1 - \frac{r}{R}\right) 2\pi r dr}{\epsilon_0}$$

$$\phi = \frac{\int dq}{\epsilon_0} = \frac{\int_0^{R/4} \sigma_0 \left(1 - \frac{r}{R}\right) 2\pi r dr}{\epsilon_0}$$

$$\therefore \frac{\phi_0}{\phi} = \frac{\sigma_0 2\pi \int_0^R \left(r - \frac{r^2}{R}\right) dr}{\sigma_0 2\pi \int_0^{R/4} \left(r - \frac{r^2}{R}\right) dr}$$

$$= \frac{\frac{R^2}{2} - \frac{R^2}{3}}{\frac{R^2}{32} - \frac{R^2}{3 \times 64}} = \frac{32}{5}$$

$$= 6.40$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)

Graph represents symmetrical distribution of speed and hence, the most probable and the average speed should be same. But the root mean square speed must be greater than the average speed.

2. Ans (B)

(A) Pb_3O_4 is insoluble in water or do not react with water.

(B) $2\text{KO}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2\text{O}_2 + \text{O}_{2(g)} \uparrow$

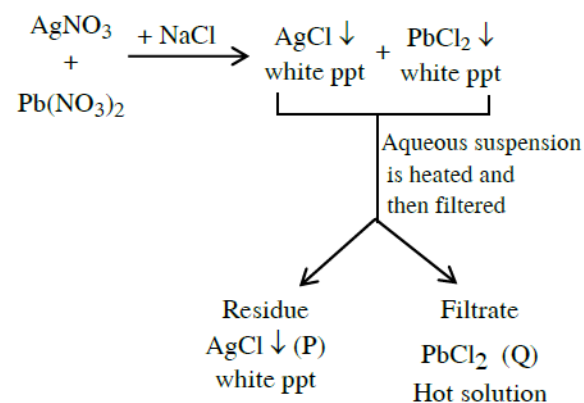
(C) $\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\text{O}_2$

(D) $\text{Li}_2\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{LiOH} + \text{H}_2\text{O}_2$

3. Ans (A)

X : Ag P : AgCl

Y : Pb Q : PbCl₂



$\text{AgCl} + 2\text{NH}_3 \text{ solution} \rightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl}$

(P) (excess) clear solution

$\text{AgCl} + 2\text{Na}_2\text{S}_2\text{O}_3 \text{ sol.} \rightarrow \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2] + \text{NaCl}$

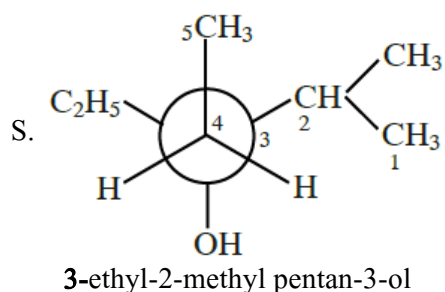
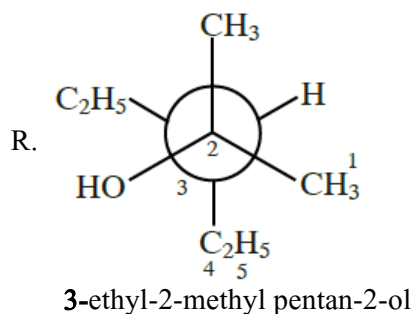
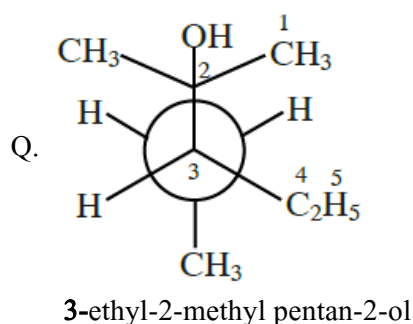
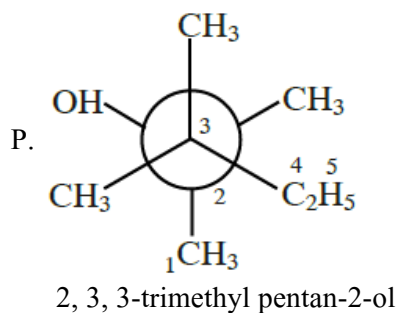
(P) (excess) clear solution

$\text{PbCl}_2 + 2\text{KI} \rightarrow \text{PbI}_2 \downarrow + 2\text{KCl}$

Hot solution (yellow ppt)

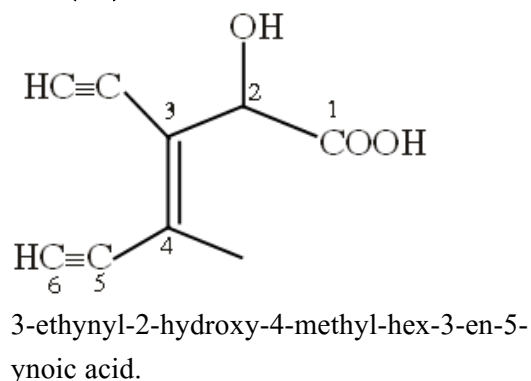
(Q)

4. Ans (C)

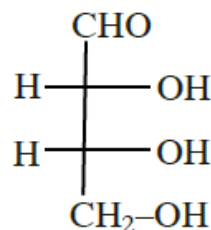


Q and R is same.

5. Ans (D)

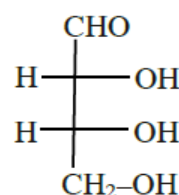


6. Ans (C)



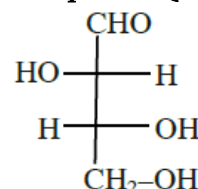
D-Erythrose

Compound P



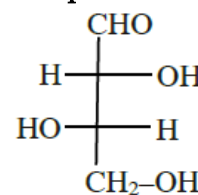
It is Identical P-2

Compound Q



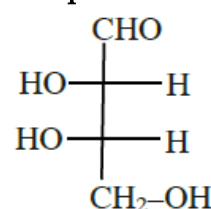
It is Diastereomer Q-1

Compound R



It is Diastereomer R-1

Compound S



It is Enantiomer S-3

Ans. P-2, Q-1, R-1, S-3

PART-2 : CHEMISTRY

SECTION-I (ii)

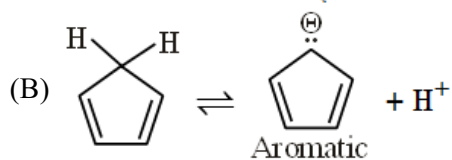
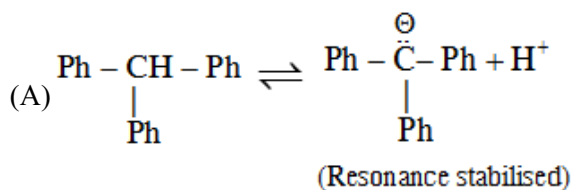
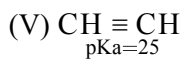
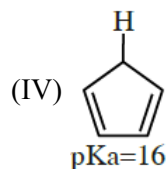
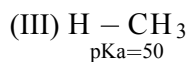
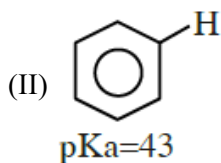
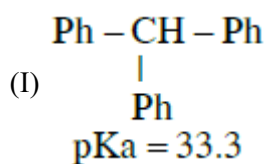
7. Ans (A,B,C)

For 1 mole Vander Waal's gas

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

If $P_{\text{ext}} = P$, means process is reversible. For Vanderwaal gas, expression is correct for all reversible process.

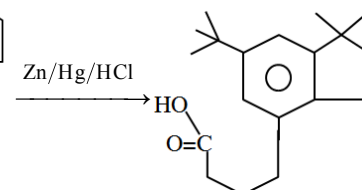
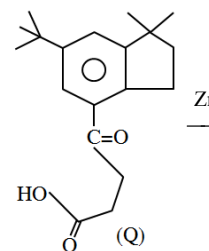
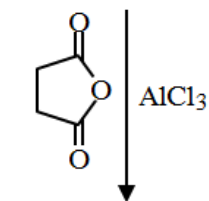
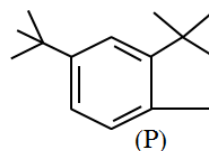
8. Ans (A,B,C)



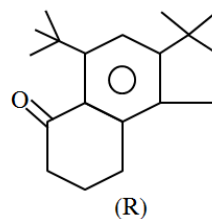
(C) $-\text{NO}_2$ is $-I$ group (electron withdrawing group). It increases acid strength.

(D) Acid strength order : $\text{IV} > \text{V} > \text{I} > \text{II} > \text{III}$

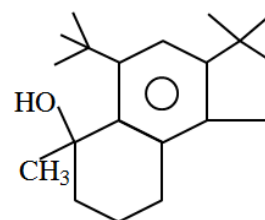
9. Ans (B,D)



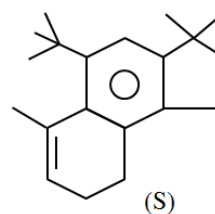
H_3PO_4



(1) CH_3MgBr
 (2) $\text{H}^+/\text{H}_2\text{O}$

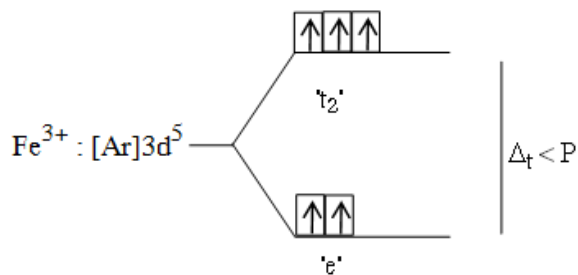


$\text{H}_2\text{SO}_4 \Delta$



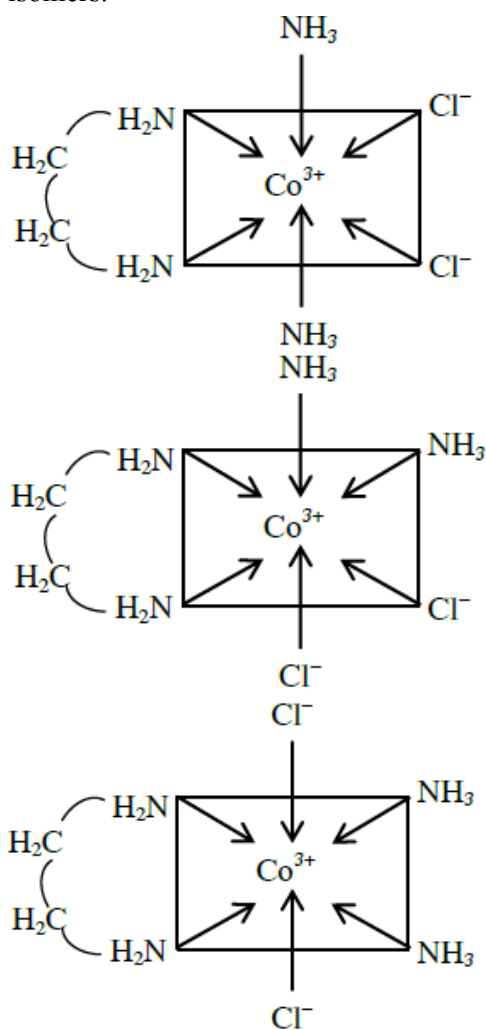
10. Ans (A,C)

(A) $[\text{FeCl}_4]^-$

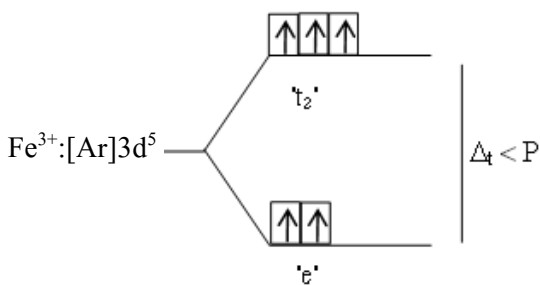


$[\text{FeCl}_4]^-$ is sp^3 hybridised and has tetrahedral geometry with 5 unpaired electrons.

(B) $[\text{Co(en)}(\text{NH}_3)_2\text{Cl}_2]^+$ has three geometrical isomers.



(C) $[\text{FeCl}_4]^-$

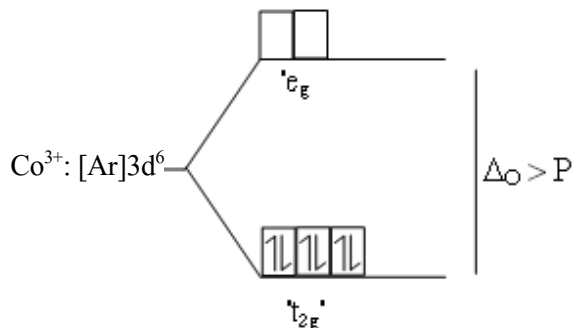


Number of unpaired electrons (n) = 5

Spin only magnetic moment

$$= \sqrt{n(n+2)} \text{ B.M.}$$

$$= 5.92 \text{ B.M. } [\text{Co(en)}(\text{NH}_3)_2\text{Cl}_2]^+$$



Number of unpaired electrons (n) = 0

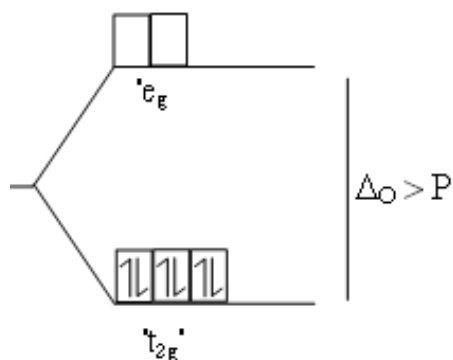
Spin only magnetic moment

$$= \sqrt{n(n+2)} \text{ B.M.}$$

$$= 0$$

(D) $[\text{Co(en)}(\text{NH}_3)_2\text{Cl}_2]^+$

$\text{Co}^{3+} : [\text{Ar}]3d^6$



$[\text{Co(en)}(\text{NH}_3)_2\text{Cl}_2]^+$ is d^2sp^3 hybridised and has octahedral geometry with 0 unpaired electron.

11. Ans (A,B,D)

Hypochlorite ion : $\text{ClO}^\ominus$

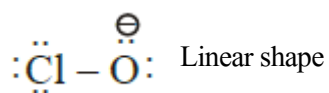
Chlorate ion : $\text{ClO}_3^\ominus$

Per chlorate ion : $\text{ClO}_4^\ominus$

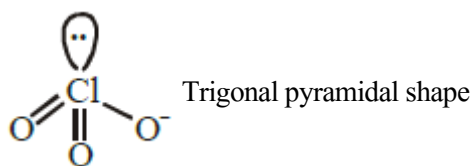
(A) Acidic order : $\text{HClO}^{+1} < \text{HClO}_3^{+5} < \text{HClO}_4^{+7}$

Conjugate base order : $\text{ClO}^- > \text{ClO}_3^- > \text{ClO}_4^-$

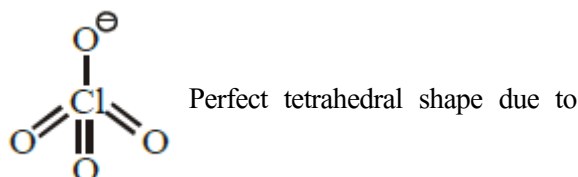
Hypochlorite ion : $(\text{ClO}^\ominus)$



Chlorate ion : $(\text{ClO}_3^\ominus)$



Per chlorate ion : $(\text{ClO}_4^\ominus)$



resonance

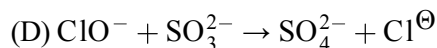
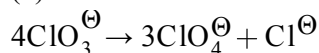
In chlorate ion bond angle changes due to presence of lone pair on chlorine atom. While hypochlorite ion is linear and perchlorate ion is tetrahedral and there is no effect of lone pair on hypochlorite ion.

(C) Disproportionation reaction of

(i) hypochlorite ion :



(ii) Chlorate ion :



12. Ans (A,C)

$$(A) Z_M = 2 \times \frac{1}{2} = 1$$

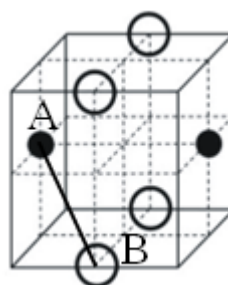
$$Z_X = 4 \times \frac{1}{4} = 1$$

$\therefore$ Empirical formula is MX

(B) Coordinate numbers of both M and X is 8.

(C) Bond length of M - X bond

$$= AB = \sqrt{3} \cdot \frac{a}{2} = 0.866a$$



(D) $r_M : r_X = (\sqrt{3} - 1) : 1 = 0.732 : 1.000$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (0.11)

No. of eq. of oxalic acid = No. of eq. of NaOH

$$\text{or } \frac{5.00 \times 0.10}{1000} \times 2 = \frac{9.0 \times M}{1000} \times 1$$

$$\therefore \text{Molarity of NaOH solution} = \frac{1}{9} = 0.11\text{M}$$

2. Ans (0.25)

$$K_{eq} = \frac{[B]}{[A]} \quad K_{1000} = \frac{10}{1} = 10 \text{ and } K_{2000} = \frac{100}{1} = 100$$

$$\begin{aligned} \text{Now, } \frac{\Delta G_{1000}^0}{\Delta G_{2000}^0} &= \frac{(-RT \ln K_{eq})_{1000}}{(-RT \ln K_{eq})_{2000}} \\ &= \frac{1000 \times \ln 10}{2000 \times \ln 100} \\ &= 0.25 \end{aligned}$$

3. Ans (13.00 to 13.60)

$$E_{\text{cell}}^0 = 1.23 - 0.00 = 1.23 \text{ V} \quad \Delta G_{\text{cell}}^0 = -nFE_{\text{cell}}^0$$

$$= -2 \times 96500 \times 1.23 \text{ J}$$

∴ Work derived from this fuel cell

$$= \frac{70}{100} \times (-\Delta G_{\text{cell}}^0) \times 10^{-3} = x \text{ J}$$

Since insulated vessel, hence $q = 0$

From equation, for monoatomic gas,

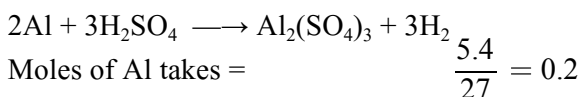
$$w = \Delta U \Rightarrow x = nC_{V,m}\Delta T \left\{ C_{V,m} = \frac{3R}{2} \right\}$$

$$\text{or, } \frac{70}{100} \times (2 \times 96500 \times 1.23) \times 10^{-3}$$

$$= 1 \times \frac{3}{2} \times 8.314 \times \Delta T$$

$$\therefore \Delta T = 13.32$$

4. Ans (6.00 to 6.20)



$$\text{moles of H}_2\text{SO}_4 \text{ taken} = \frac{50 \times 5.0}{1000} = 0.25$$

$$\text{As } \frac{0.2}{2} > \frac{0.25}{3}, \text{H}_2\text{SO}_4 \text{ is limiting reagent}$$

$$\text{Now, moles of H}_2 \text{ formed} = \frac{3}{3} \times 0.25 = 0.25$$

$$\therefore \text{Volume of H}_2 \text{ gas formed} = \frac{nRT}{P}$$

$$= \frac{0.25 \times 0.082 \times 300}{1} = 6.15 \text{ L}$$

5. Ans (1.19 to 1.21)

$$\text{Initial moles of U}^{238} = \frac{68 \times 10^{-6}}{238} = x$$

$$\text{Moles of U}^{238} \text{ decayed in three half-lives} = \frac{7}{8}x$$

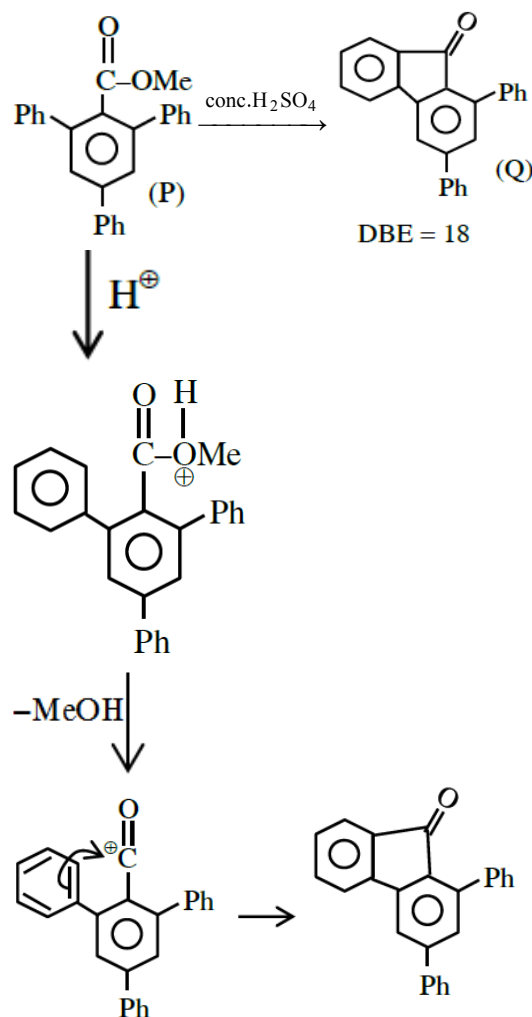
In decay from U^{238} to Pb^{206} , each U^{238} atom decays and produces 8 α -particles and hence, total number of α -particles emitted out

$$= \left(\frac{7}{8}x \right) \times 8 \times N_A$$

$$= 7 \times \frac{68 \times 10^{-6}}{238} \times 6.022 \times 10^{23}$$

$$= 1.204 \times 10^{18}$$

6. Ans (18.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

$$x^2 + 20x - 2020 = 0 \text{ has two roots } a, b \in \mathbb{R}$$

$$x^2 - 20x + 2020 = 0 \text{ has two roots } c, d \in \text{complex}$$

$$ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$$

$$= a^2c - ac^2 + a^2d - ad^2 + b^2c - bc^2 + b^2d - bd^2$$

$$= a^2(c+d) + b^2(c+d) - c^2(a+b) - d^2(a+b)$$

$$= (c+d)(a^2+b^2) - (a+b)(c^2+d^2)$$

$$= (c+d)((a+b)^2 - 2ab) - (a+b)((c+d)^2 - 2cd)$$

$$= 20[(20)^2 + 4040] + 20[(20)^2 - 4040]$$

$$= 20[(20)^2 + 4040 + (20)^2 - 4040]$$

$$= 20 \times 800 = 16000$$

2. **Ans (C)**

$f(x)$ is a non-periodic, continuous and odd function

$$f(x) = \begin{cases} -x^2 + x \sin x, & x < 0 \\ x^2 - x \sin x, & x \geq 0 \end{cases}$$

$$f(-\infty) = \lim_{x \rightarrow -\infty} (-x^2) \left(1 - \frac{\sin x}{x}\right) = -\infty$$

$$f(\infty) = \lim_{x \rightarrow \infty} x^2 \left(1 - \frac{\sin x}{x}\right) = \infty$$

$\Rightarrow$ Range of $f(x) = \mathbb{R}$

$\Rightarrow f(x)$ is an onto function ... (1)

$$f'(x) = \begin{cases} -2x + \sin x + x \cos x, & x < 0 \\ 2x - \sin x - x \cos x, & x \geq 0 \end{cases}$$

For $(0, \infty)$

$$f'(x) = (x - \sin x) + x(1 - \cos x)$$

always +ve or 0 always +ve or 0

$$\Rightarrow f'(x) > 0$$

$$\Rightarrow f'(x) \geq 0, \forall x \in (-\infty, \infty)$$

equality at $x = 0$

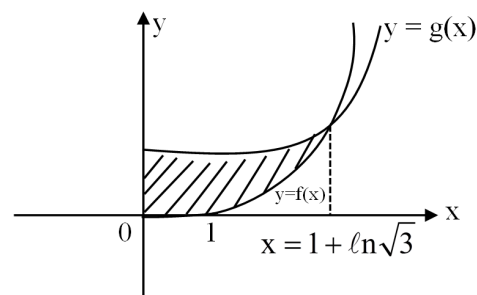
$\Rightarrow f(x)$ is one-one function ... (2)

From (1) & (2), $f(x)$ is both one-one & onto.

3. **Ans (A)**

$$\text{Here } f(x) = \begin{cases} 0 & x \leq 1 \\ e^{x-1} - e^{1-x} & x \geq 1 \end{cases}$$

$$\& g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$$



$$\text{solve } f(x) \& g(x) \Rightarrow x = 1 + \ln \sqrt{3}$$

So bounded area

$$= \int_0^1 \frac{1}{2}(e^{x-1} + e^{1-x}) dx +$$

$$\int_1^{1+\ln \sqrt{3}} \left[\frac{1}{2}(e^{x-1} + e^{1-x}) - (e^{x-1} - e^{1-x}) \right] dx$$

$$= \frac{1}{2} [e^{x-1} - e^{1-x}]_0^1 + \left[-\frac{1}{2}e^{x-1} - \frac{3}{2}e^{1-x} \right]_1^{1+\ln \sqrt{3}}$$

$$= \frac{1}{2} \left[e - \frac{1}{e} \right] + \left[\left(-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right) + 2 \right] = 2 - \sqrt{3} + \frac{1}{2} \left(e - \frac{1}{e} \right)$$

4. **Ans (A)**

$$y^2 = 4\lambda x, P(\lambda, 2\lambda)$$

Slope of the tangent to the parabola at point P

$$\frac{dy}{dx} = \frac{4\lambda}{2y} = \frac{4\lambda}{2 \cdot 2\lambda} = 1$$

Slope of the tangent to the ellipse at P

$$\frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$$

As tangents are perpendicular $y' = -1$

$$\Rightarrow \frac{2\lambda}{a^2} - \frac{4\lambda}{b^2} = 0 \Rightarrow \frac{a^2}{b^2} = \frac{1}{2}$$

$$\Rightarrow e = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}$$

5. **Ans (B)**

$$P(H) = \frac{2}{3} \text{ for } C_1$$

$$P(H) = \frac{1}{3} \text{ for } C_2$$

for C_1

No. of Heads (α)	0	1	2
Probability	$\frac{1}{9}$	$\frac{4}{9}$	$\frac{4}{9}$

for C_2

No. of Heads (β)	0	1	2
Probability	$\frac{4}{9}$	$\frac{4}{9}$	$\frac{1}{9}$

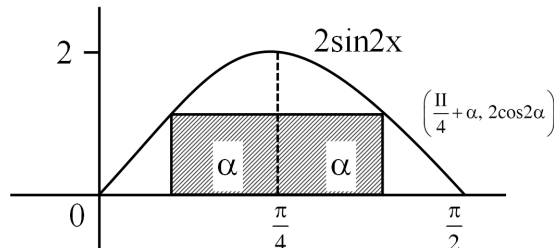
for real and equal roots

$$\alpha^2 = 4\beta$$

$$(\alpha, \beta) = (0, 0), (2, 1)$$

$$\text{So, probability} = \frac{1}{9} \times \frac{4}{9} + \frac{4}{9} \times \frac{4}{9} = \frac{20}{81}$$

6. Ans (C)



$$\text{Perimeter} = 2(2\alpha + 2 \cos 2\alpha)$$

$$P = 4(\alpha + \cos 2\alpha)$$

$$\frac{dP}{d\alpha} = 4(1 - 2 \sin 2\alpha) = 0$$

$$\sin 2\alpha = \frac{1}{2}$$

$$2\alpha = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\frac{d^2P}{d\alpha^2} = -4 \cos 2\alpha$$

$$\text{for maximum } \alpha = \frac{\pi}{12}$$

$$\text{Area} = (2\alpha) (2 \cos 2\alpha)$$

$$= \frac{\pi}{6} \times 2 \times \frac{\sqrt{3}}{2} = \frac{\pi}{2\sqrt{3}}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,C)

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = (x^2 + \sin x)(x-1)$$

$$f(1^+) = f(1^-) = f(1) = 0$$

$$fg(x) : f(x).g(x) \quad fg : \mathbb{R} \rightarrow \mathbb{R}$$

$$\text{let } fg(x) = h(x) = f(x).g(x) \quad h : \mathbb{R} \rightarrow \mathbb{R}$$

$$\text{option (c)} \quad h'(x) = f'(x)g(x) + f(x)g'(x)$$

$$h'(1) = f'(1)g(1) + 0,$$

$$(\text{as } f(1) = 0, g'(x) \text{ exists})$$

$\Rightarrow$ if $g(x)$ is differentiable then $h(x)$ is also differentiable (true)

option (A) If $g(x)$ is continuous at $x = 1$ then

$$g(1^+) = g(1^-) = g(1)$$

$$h'(1^+) = \lim_{h \rightarrow 0^+} \frac{h(1+h) - h(1)}{h}$$

$$h'(1^+) = \lim_{h \rightarrow 0^+} \frac{f(1+h)g(1+h) - 0}{h} = f'(1)g(1)$$

$$h'(1^-) = \lim_{h \rightarrow 0^+} \frac{f(1-h)g(1-h) - 0}{-h} = f'(1)g(1)$$

So $h(x) = f(x).g(x)$ is differentiable

at $x = 1$ (True)

$$\text{option (B) (D)} \quad h'(1^+) = \lim_{h \rightarrow 0^+} \frac{h(1+h) - h(1)}{-h}$$

$$h'(1^+) = \lim_{h \rightarrow 0^+} \frac{f(1+h)g(1+h)}{h} = f'(1)g(1^+)$$

$$h'(1^-) = \lim_{h \rightarrow 0^+} \frac{f(1-h)g(1-h)}{-h} = f'(1).g(1^-)$$

$$\Rightarrow g(1^+) = g(1^-)$$

So we cannot comment on the continuity and differentiability of the function.

8. Ans (B,C,D)

$$\det(M) \neq 0$$

$$M^{-1} = \text{adj}(\text{adj } M)$$

$$M^{-1} = \det(M).M$$

$$M^{-1}M = \det(M).M^2$$

$$I = \det(M).M^2 \quad \dots(i)$$

$$\det(I) = (\det(M))^5$$

$$1 = \det(M) \quad \dots(ii)$$

$$\text{From (i)} \quad I = M^2$$

$$(\text{adj } M)^2 = \text{adj}(M^2) = \text{adj } I = I$$

9. Ans (B,C)

$$|z^2 + z + 1| = 1$$

$$\Rightarrow \left| \left(z + \frac{1}{2} \right)^2 + \frac{3}{4} \right| = 1$$

$$\Rightarrow \left| \left(z + \frac{1}{2} \right)^2 + \frac{3}{4} \right| \leq \left| z + \frac{1}{2} \right|^2 + \frac{3}{4}$$

$$\Rightarrow 1 \leq \left| z + \frac{1}{2} \right|^2 + \frac{3}{4} \Rightarrow \left| \left(z + \frac{1}{2} \right)^2 \right| \geq \frac{1}{4}$$

$$\Rightarrow \left| z + \frac{1}{2} \right| \geq \frac{1}{2}$$

$$\text{also } |(z^2 + z) + 1| = 1 \geq ||z^2 + z| - 1|$$

$$\Rightarrow |z^2 + z| - 1 \leq 1$$

$$\Rightarrow |z^2 + z| \leq 2$$

$$\Rightarrow ||z|^2 - |z|| \leq |z^2 + z| \leq 2$$

$$\Rightarrow |r^2 - r| \leq 2$$

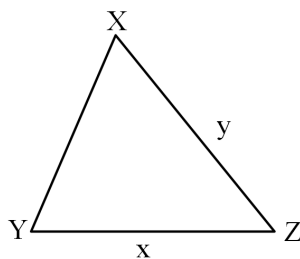
$$\Rightarrow r = |z| \leq 2; \forall z \in S$$

Also we can always find root of the equation z^2

$$+ z + 1 = e^{i\theta}; \forall \theta \in \mathbb{R}$$

Hence set 'S' is infinite

10. Ans (B,C)



$$\tan \frac{X}{2} + \tan \frac{Z}{2} = \frac{2y}{x+y+z}$$

$$\frac{\Delta}{S(S-x)} + \frac{\Delta}{S(S-z)} = \frac{2y}{2S}$$

$$\frac{\Delta}{S} \left(\frac{2S - (x+z)}{(S-x)(S-z)} \right) = \frac{y}{S}$$

$$\Rightarrow \frac{\Delta y}{S(S-x)(S-z)} = \frac{y}{S}$$

$$\Rightarrow \Delta^2 = (S-x)^2 (S-z)^2$$

$$\Rightarrow S(S-y) = (S-x)(S-z)$$

$$\Rightarrow (x+y+z)(x+z-y) = (y+z-x)(x+y-z)$$

$$\Rightarrow (x+z)^2 - y^2 = y^2 - (z-x)^2$$

$$\Rightarrow (x+z)^2 + (x-z)^2 = 2y^2$$

$$\Rightarrow x^2 + z^2 = y^2 \Rightarrow \angle Y = \frac{\pi}{2}$$

$$\Rightarrow \angle Y = \angle X + \angle Z$$

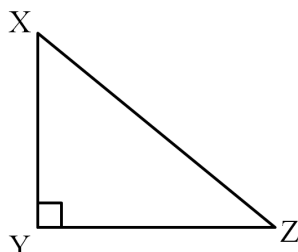
$$\tan \frac{X}{2} = \frac{\Delta}{S(S-x)}$$

$$\tan \frac{X}{2} = \frac{\frac{1}{2}xz}{\frac{(y+z)^2 - x^2}{4}}$$

$$\tan \frac{X}{2} = \frac{2xz}{y^2 + z^2 + 2yz - x^2}$$

$$\tan \frac{X}{2} = \frac{2xz}{2z^2 + 2yz} \quad (\text{using } y^2 = x^2 + z^2)$$

$$\tan \frac{X}{2} = \frac{x}{y+z}$$



11. Ans (A,B)

Point of intersection of L_1 & L_2 is $(1, 0, 1)$

Line L passes through $(1, 0, 1)$

$$\frac{1-\alpha}{\ell} = -\frac{1}{m} = \frac{1-\gamma}{-2} \quad \dots(1)$$

acute angle bisector of L_1 & L_2

$$\vec{r} = \hat{i} + \hat{k} + \lambda \left(\frac{\hat{i} - \hat{j} + 3\hat{k} - 3\hat{i} - \hat{j} + \hat{k}}{\sqrt{11}} \right)$$

$$\vec{r} = \hat{i} + \hat{k} + t(\hat{i} + \hat{j} - 2\hat{k})$$

$$\Rightarrow \frac{\ell}{1} = \frac{m}{1} = \frac{-2}{-2} \Rightarrow \ell = m = 1$$

$$\text{From (1)} \quad \frac{1-\alpha}{1} = -1 \Rightarrow \alpha = 2$$

$$\& \frac{1-\gamma}{-2} = -1 \Rightarrow \gamma = -1$$

12. Ans (A,B,D)

$$(A) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x \geq 1 - \frac{x^2}{2}$$

$$\int_0^1 x \cos x \geq \int_0^1 x \left(1 - \frac{x^2}{2}\right) = \frac{1}{2} - \frac{1}{8}$$

$$\int_0^1 x \cos x \geq \frac{3}{8} \quad (\text{True})$$

$$(B) \sin x \geq x - \frac{x^3}{6}$$

$$\int_0^1 x \sin x \geq \int_0^1 x \left(x - \frac{x^3}{6}\right) dx$$

$$\int_0^1 x \sin x \geq \frac{1}{3} - \frac{1}{30} \Rightarrow \int_0^1 x \sin x \geq \frac{3}{8} \quad (\text{True})$$

$$(D) \int_0^1 x^2 \sin x \geq \int_0^1 x^2 \left(x - \frac{x^3}{6}\right) dx$$

$$\int_0^1 x^2 \sin x \geq \frac{1}{4} - \frac{1}{36}$$

$$\int_0^1 x^2 \sin x \geq \frac{2}{9} \quad (\text{True})$$

$$(C) \cos x < 1$$

$$x^2 \cos x < x^2$$

$$\int_0^1 x^2 \cos x < \int_0^1 x^2 dx$$

$$\int_0^1 x^2 \cos x < \frac{1}{3}$$

So option 'C' is incorrect.

PART-3 : MATHEMATICS

SECTION-II

1. Ans (8.00)

$$\frac{3^{y_1} + 3^{y_2} + 3^{y_3}}{3} \geq \left[3^{(y_1+y_2+y_3)}\right]^{\frac{1}{3}}$$

$$\Rightarrow 3^{y_1} + 3^{y_2} + 3^{y_3} \geq 3^4$$

$$\Rightarrow \log_3(3^{y_1} + 3^{y_2} + 3^{y_3}) \geq 4$$

$$\Rightarrow m = 4$$

$$\text{Also, } \frac{x_1 + x_2 + x_3}{3} \geq \sqrt[3]{x_1 x_2 x_3}$$

$$\Rightarrow x_1 x_2 x_3 \leq 27$$

$$\Rightarrow \log_3 x_1 + \log_3 x_2 + \log_3 x_3 \leq 3$$

$$\Rightarrow M = 3$$

$$\text{Thus, } \log_2(m^3) + \log_3(M^2) = 6 + 2$$

$$= 8$$

2. Ans (1.00)

Given

$$2(a_1 + a_2 + \dots + a_n) = b_1 + b_2 + \dots + b_n$$

$$\Rightarrow 2 \times \frac{n}{2} (2c + (n-2)x_2) = c \left(\frac{2^n - 1}{2 - 1}\right)$$

$$\Rightarrow 2n^2 - 2n = c(2^n - 1 - 2n)$$

$$\Rightarrow c = \frac{2n^2 - 2n}{2^n - 1 - 2n} \in \mathbb{N}$$

$$\text{So, } 2n^2 - 2n \geq 2^n - 1 - 2n$$

$$\Rightarrow 2n^2 + 1 \geq 2^n \Rightarrow n < 7$$

$$\Rightarrow n \text{ can be } 1, 2, 3, \dots,$$

Checking c against these values of n

we get c = 12 (when n = 3)

Hence number of such c = 1

3. Ans (1.00)

$$\text{Let } \pi x - \frac{\pi}{4} = \theta \in \left[\frac{-\pi}{4}, \frac{7\pi}{4}\right]$$

$$\text{So, } \left(3 - \sin\left(\frac{\pi}{2} + 2\theta\right)\right) \sin \theta \geq \sin(\pi + 3\theta)$$

$$\Rightarrow (3 - \cos 2\theta) \sin \theta \geq -\sin 3\theta$$

$$\Rightarrow \sin \theta [3 - 4 \sin^2 \theta + 3 - \cos 2\theta] \geq 0$$

$$\Rightarrow \sin \theta (6 - 2(1 - \cos 2\theta) - \cos 2\theta) \geq 0$$

$$\Rightarrow \sin \theta (4 + \cos 2\theta) \geq 0$$

$$\Rightarrow \sin \theta \geq 0$$

$$\Rightarrow \theta \in [0, \pi] \Rightarrow 0 \leq \pi x - \frac{\pi}{4} \leq \pi$$

$$\Rightarrow x \in \left[\frac{1}{4}, \frac{5}{4}\right]$$

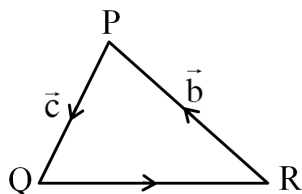
$$\Rightarrow \beta - \alpha = 1$$

4. **Ans (108.00)**

$$\text{We have } \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{c} = -\vec{a} - \vec{b}$$

Now,



$$\frac{\vec{a} \cdot (-\vec{a} - 2\vec{b})}{(-\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})} = \frac{3}{7}$$

$$\Rightarrow \frac{9 + 2\vec{a} \cdot \vec{b}}{9 - 16} = \frac{3}{7}$$

$$\Rightarrow \vec{a} \cdot \vec{b} = -6$$

$$\Rightarrow |\vec{a} \times \vec{b}|^2 = a^2 b^2 - (\vec{a} \cdot \vec{b})^2 = 9 \times 16 - 36 = 108$$

5. **Ans (5.00)**

$$f(x) = (x^2 - 1)^2 h(x); h(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$$

$$\text{Now, } f(1) = f(-1) = 0$$

$$\Rightarrow f'(\alpha) = 0, \alpha \in (-1, 1) \quad [\text{Rolle's Theorem}]$$

Also, $f'(1) = f'(-1) = 0 \Rightarrow f'(x) = 0$ has atleast 3 root, $-1, \alpha, 1$ with $-1 < \alpha < 1$

$\Rightarrow f''(x) = 0$ will have atleast 2 root, say β, γ such that

$$-1 < \beta < \alpha < \gamma < 1 \quad [\text{Rolle's Theorem}]$$

$$\text{So, } \min(m_{f''}) = 2$$

$$\text{and we find } (m_{f'} + m_{f''}) = 5 \text{ for } f(x) = (x^2 - 1)^2 h(x)$$

Thus, Ans. 5

6. **Ans (1.00)**

$$\lim_{x \rightarrow 0^+} \frac{e^{\left(\frac{\ln(1-x)}{x}\right)} - \frac{1}{e}}{x^a}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{e} \frac{e^{\left(1 + \frac{\ln(1-x)}{x}\right)} - 1}{x^a}$$

$$= \frac{1}{e} \lim_{x \rightarrow 0^+} \frac{1 + \frac{\ln(1-x)}{x}}{x^a}$$

$$= \frac{1}{e} \lim_{x \rightarrow 0^+} \frac{\ln(1-x) + x}{x^{a+1}}$$

$$= \frac{1}{e} \lim_{x \rightarrow 0^+} \frac{\left(-x - \frac{x^2}{2} - \frac{x^3}{3} - \dots\right) + x}{x^{a+1}}$$

Thus, $a = 1$

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,D	A,D	A,C,D	A,C	B,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.05	0.26 to 0.28	1.30	0.23 to 0.25	55.00	8.32 to 8.34
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	9	6	4	2	6

PART-2 : CHEMISTRY

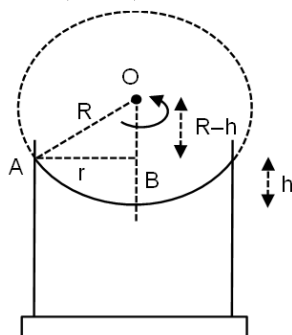
SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C	A	A,B,C,D	A,B	C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.80 to 3.20	0.20	-2640.00--2620.00 or -5280.00--5240.00	18.60	935.00	0.20
SECTION-III	Q.	1	2	3	4	5	6
	A.	9	6	6	4	5	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	A,D	A,B,D	A,B,C	A,C	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	495.00	1080.00	8.00	19.00	4.00	0.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	6	2	5	4	8

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A,D)


 In $\triangle OAB$

$$R^2 = (R - h)^2 + r^2$$

$$R^2 = R^2 - 2hR + h^2 + r^2$$

$$\Rightarrow 2hR = h^2 + r^2$$

$$\Rightarrow R = \frac{h^2 + r^2}{2h}$$

Now considering equation of surface

$$y = y_0 + \frac{\omega^2 r^2}{2g}$$

$$h = \frac{\omega^2 r^2}{2g}$$

$$\text{Now using : } \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\Rightarrow \frac{1}{v} + \frac{4}{3(H-h)} = \frac{1 - 4/3}{-R}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{3R} - \frac{4}{3H}$$

$$\Rightarrow \frac{1}{v} = \frac{2h}{3r^2} - \frac{4}{3H}$$

$$\Rightarrow \frac{1}{v} = -\frac{4}{3H} \left[1 - \frac{\omega^2 H}{4g} \right]$$

$$\Rightarrow v = \frac{3H}{4} \left[1 + \frac{\omega^2 H}{4g} \right]^{-1}$$

2. **Ans (A,D)**

By the energy conservation (ME) between bottom point and point Y

$$\frac{1}{2}mv_0^2 = mgh + \frac{1}{2}mv_1^2$$

$$\therefore v_1^2 = v_0^2 - 2gh \quad \dots (i)$$

Now at point Y the centripetal force provided by the component of mg

$$\therefore mg \sin 30^\circ = \frac{mv_1^2}{R}$$

$$\therefore v_1^2 = \frac{gR}{2}$$

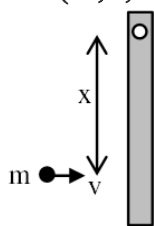
$\therefore$ from (i)

$$\frac{gR}{2} = v_0^2 - 2gh$$

At point x and z of circular path, the points are at same height but less than h. So the velocity more than a point y.

So required centripetal = $\frac{mv^2}{r}$ is more.

3. **Ans (A,C,D)**



by the angular momentum conservation about the suspension point.

$$mvx = \left(\frac{m\ell^2}{3} + mx^2 \right) \omega$$

$$\therefore \omega = \frac{mvx}{\frac{m\ell^2}{3} + mx^2} = \frac{2vx}{\ell^2 + 3x}$$

$$\text{For maximum } \omega \Rightarrow \frac{d\omega}{dx} = 0$$

$$\therefore x_M = \frac{\ell}{\sqrt{3}}$$

$$\text{So the } \omega = \frac{V}{2\ell} \sqrt{3}$$

4. **Ans (A,C)**

$$\lambda_{\min} = \frac{hc}{eV}$$

$$\Rightarrow \lambda_{\min} \propto \frac{1}{V} \Rightarrow (\lambda_{\min})_{\text{new}} = \frac{\lambda_2}{2}$$

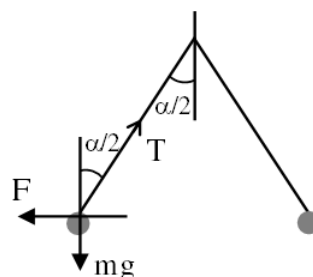
$$\therefore I = \frac{dN}{dt} \times \frac{hc}{\lambda}$$

$$\therefore \frac{dN}{dt} \text{ decreases}$$

Hence I decreases

5. **Ans (B,C)**

The net electric force on any sphere is lesser but by Coulomb law the force due to one sphere to another remain the same.



In equilibrium

$$T \cos \frac{\alpha}{2} = mg$$

$$\text{and } T \sin \frac{\alpha}{2} = F$$

After immersed in dielectric liquid.

As given no change in angle α .

$$\text{So } T \cos \frac{\alpha}{2} = mg - V\rho g$$

when $\rho = 800 \text{ Kg/m}^3$

$$\text{and } T \sin \frac{\alpha}{2} = \frac{F}{e_r}$$

$$\therefore \frac{mg}{F} = \frac{mg - V\rho g}{\frac{F}{e_r}}$$

$$\frac{1}{e_r} = 1 - \frac{\rho}{d}$$

d = density of sphere

$$\frac{1}{21} = 1 - \frac{800}{d}$$

$$d = 840$$

6. **Ans (A,B,C,D)**



$$\text{at } t = 0, \left. \begin{matrix} x = 0, y = 0 \\ u = 1 \end{matrix} \right\} \text{ given } y = \frac{x^2}{2}$$

$$\frac{dy}{dt} = \frac{1}{2} \cdot 2x \frac{dx}{dt}$$

$$\Rightarrow v_y = xv_x$$

$$\text{difference wrt time } a_y = \frac{dx}{dt} \cdot V_x + xa_x$$

$$a_y = v_x^2 + xa_x$$

Option

(A) If $a_x = 1$ and particle is at origin

$$(x = 0, y = 0)$$

$$a_y = v_x^2$$

$$a_y = 1^2 = 1$$

At origin, at $t = 0$ sec

speed = 1 given

(B) Option

$$a_y = v_x^2 + x a_x$$

given in option B, $a_x = 0$

$$\Rightarrow a_y = v_x^2$$

If $a_x = 0$, $v_x = \text{constant} = 1$, (all the time)

$$\Rightarrow a_y = 1^2 = 1 \text{ (all the time)}$$

(C) at $t = 0$, $x = 0$ $v_y = x v_x$

speed = 1

$$v_y = 0$$

$$v_x = 1$$

(D) $a_y = v_x^2 + x a_x$

$$v_y = x v_x$$

$a_x = 0$ (given in D option)

$$\Rightarrow a_y = v_x^2$$

If $a_x = 0 \Rightarrow v_x = \text{constant initially}$ ($v_x = 1$)

$$\Rightarrow a_y = 1^2 = 1$$

at $t = 1$ sec

$$v_y = 0 + a_y \times t = 1 \times 1 = 1$$

$$\tan \theta = \frac{v_y}{v_x} = 1$$

($\theta \rightarrow$ angle with x axis)

$$\tan \theta = \frac{v_y}{v_x} = \frac{1}{1} = 1$$

$$\theta = 45^\circ$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.05)**

$$W = (\Delta P)_{\text{avg}} \times 4\pi R^2 a$$

$$\simeq \left| \frac{dP}{2} \cdot 4\pi R^2 a \right|$$

{for small change $(\Delta P)_{\text{avg}} < P >$ arithmetic mean}

$$= P V^\gamma = c \Rightarrow dP = -\gamma \frac{P}{V} dV = -\frac{\gamma P_0}{V} 4\pi R^2 a$$

$$= \frac{\gamma P_0}{2V} \times 4\pi R^2 a \times 4\pi R^2 a$$

$$= \frac{\gamma P_0}{2 \times 4\pi R^3} 4\pi R^2 a \times 4\pi R^2 a$$

$$= (4\pi R P \times a^2) \frac{3\gamma}{2}$$

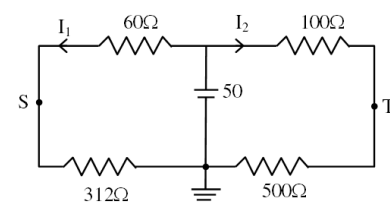
$$\therefore x \simeq 2.05$$

2. **Ans (0.26 to 0.28)**

$$R'_3 = 300 (1 + \alpha \Delta T)$$

$$= 312 \Omega$$

Now



$$I_1 = \frac{50}{372} \text{ and } I_2 = \frac{50}{600}$$

$$V_S - V_T = 312I_1 - 500I_2$$

$$= 41.94 - 41.67$$

$$= 0.27 \text{ V}$$

3. **Ans (1.30)**

$$U = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} V^2$$

$$\text{Let } C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$$

$$\frac{1}{C_{\text{eq}} \pm \Delta C_{\text{eq}}} = \frac{1}{C_1 \pm \Delta C_1} + \frac{1}{C_2 \pm \Delta C_2}$$

$$\Rightarrow C_{\text{eq}} \pm \Delta C_{\text{eq}} \simeq \frac{C_1 C_2 + C_1 \Delta C_2 + C_2 \Delta C_1}{C_1 + C_2 + \Delta C_1 + \Delta C_2}$$

$$= \frac{1200 \left(1 \pm \frac{12}{1200} \right)}{\left(1 \pm \frac{25}{5000} \right)}$$

$$= 1200 \left[1 \pm \left(\frac{1}{100} - \frac{1}{200} \right) \right]$$

$$\frac{\Delta U}{U} \times 100 = \frac{\Delta C_{\text{eq}}}{C_{\text{eq}}} \times 100 + \frac{2\Delta V}{V} \times 100$$

$$= \frac{1}{200} \times 100 + 2 \times \frac{0.02}{5} \times 100 = 1.3 \%$$

4. Ans (0.23 to 0.25)

$$\frac{dV}{V} = \frac{3da}{a}$$

$$B = -V \frac{dP}{dV} = \frac{-V(\rho gh)}{dV} = \frac{-\rho gh}{3da} a$$

$$70 \times 10^9 = \frac{1 \times 5000 \times 10^3 \times 10 \times 1}{3 \times da}$$

$$da = \Delta a = \frac{5}{21} \times 10^{-2} \text{m} = 2.38 \text{mm}$$

5. Ans (55.00)

Mutual inductance is producing flux in same direction as self inductance.

$$\therefore U = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2$$

$$\Rightarrow U = \frac{1}{2} \times (10 \times 10^{-3})^2 + \frac{1}{2} \times (20 \times 10^{-3}) \times 2^2$$

$$+ (5 \times 10^{-3}) \times 1 \times 2$$

$$= 55 \text{ mJ}$$

6. Ans (8.32 to 8.34)

$$\frac{dQ}{dt} = \sigma e A (T^4 - T_0^4) \quad \dots(i)$$

$$\frac{dQ}{A dt} = e \sigma (T_0 + \Delta T)^4 - T_0^4 = \sigma T_0^4 \left[\left(1 + \frac{\Delta T}{T_0} \right)^4 - 1 \right]$$

$$= e \sigma T_0^4 \left[\left(1 + 4 \frac{\Delta T}{T_0} \right) - 1 \right]$$

$$\frac{dQ}{A dt} = \sigma e T_0^3 \cdot 4 \Delta T \quad \dots(ii)$$

Now from equ. (i)

$$ms \frac{dT}{dt} = \sigma e A (T^4 - T_0^4)$$

$$\frac{dT}{dt} = \frac{\sigma e A}{ms} [(T_0 + \Delta T)^4 - T_0^4]$$

$$= \frac{\sigma e A}{ms} T_0^4 \times \left[\left(1 + \frac{\Delta T}{T_0} \right)^4 - 1 \right]$$

$$\frac{dT}{dt} = \frac{\sigma e A}{ms} T_0^4 \cdot 4 \Delta T$$

$$\frac{dT}{dt} = e \Delta T; \left(K = \frac{4 \sigma e A T_0^3}{ms} \right)$$

$$\Rightarrow 4 \sigma e A T_0^3 = \frac{K}{A} (ms)$$

from equ. (i)

$$\frac{dQ}{A dt} = e \sigma T_0^3 \cdot 4 \Delta T$$

$$700 = (K/A) (ms) \Delta T$$

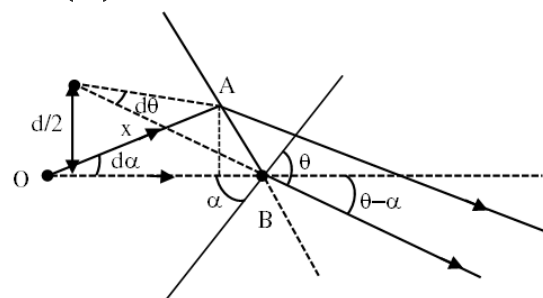
$$\therefore \Delta T = \frac{700 \times 5 \times 10^{-2}}{10^{-3} \times 4200} = \frac{50}{6} = \frac{25}{3}$$

$$\Delta T = 8.33$$

PART-1 : PHYSICS

SECTION-III

1. Ans (2)



We will assume that observer sees the image of object through edge $\Rightarrow \alpha = 45^\circ$

$$AB = \frac{12da}{\cos \alpha} = \frac{xd\theta}{\cos \theta}$$

By applying Snell's Law

$$\frac{4}{3} \sin \alpha = 1 \sin \theta$$

$$\frac{4}{3} \cos \alpha d\alpha = \cos \theta d\theta$$

$$\Rightarrow \frac{9}{\cos^2 \alpha} = \frac{x}{\cos^2 \theta}$$

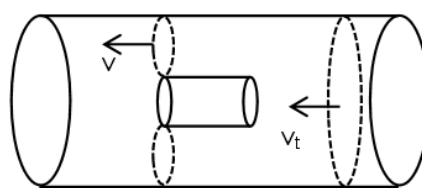
$$1 \sin \theta = \frac{4}{3} \sin \alpha$$

$$\Rightarrow \sin \theta = \frac{2\sqrt{2}}{3} \Rightarrow x = 18 \times \frac{1}{9} = 2$$

$$d = 2x \sin (\theta - \alpha)$$

$$= 4 \times \frac{1}{\sqrt{2}} \left(\frac{2\sqrt{2}}{3} - \frac{1}{3} \right) = \frac{8 - 2\sqrt{2}}{3} \approx 1.73 \approx 2$$

2. Ans (9)



with respect to train

Applying Bernoulli's equation

$$P_0 + \frac{1}{2} \rho v_t^2 = P + \frac{1}{2} \rho v^2$$

$$P_0 - P = \frac{1}{2} \rho (v^2 - v_t^2) \quad \dots (i)$$

From equation of continuity

$$\text{Also, } 4S_t v_t = v \times 3S_t \Rightarrow v = \frac{4}{3} v_t \quad \dots (ii)$$

From (i) and (ii)

$$P_0 - P = \frac{1}{2} \rho \left(\frac{16}{9} v_t^2 - v_t^2 \right) = \frac{1}{2} \rho \frac{7v_t^2}{9}$$

$$\therefore N = 9$$

3. **Ans (6)**

$$E = \frac{V}{d} = \frac{200}{0.01} = 2 \times 10^4 \text{ V/m}$$

When terminal velocity is achieved

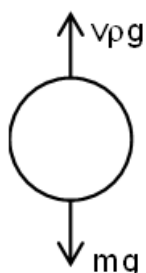
$$qE = mg$$

$$\Rightarrow n \times 1.6 \times 10^{-19} \times 2 \times 10^4$$

$$= \frac{4\pi}{3} (8 \times 10^{-7})^3 \times 900 \times 10$$

$$\Rightarrow n \approx 6$$

4. **Ans (4)**



$$480 \times g = v_p \rho_1 g$$

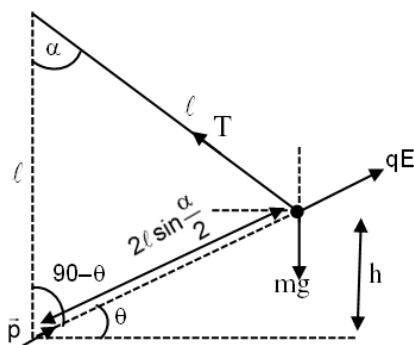
$$(480 - N) g = v_p \rho_2 g$$

$$\frac{480 - N}{480} = \frac{\rho_2}{\rho_1}$$

$$\left(1 - \frac{N}{480}\right) = \frac{e^{-h_2/h_0}}{e^{-h_1/h_0}} = e^{\frac{h_1 - h_2}{h_0}} = e^{-\frac{50}{6000}}$$

$$1 - \frac{N}{480} = 1 - \frac{50}{6000} \Rightarrow N = \frac{50 \times 480}{6000} = 4$$

5. **Ans (2)**



$$U_i = 0$$

$$U_f = \frac{kqP}{(2l \sin \frac{\alpha}{2})^2} + mgh \quad \dots (i)$$

Now, from ΔOAB

$$\alpha + 90 - \theta + 90 - \theta = 180$$

$$\Rightarrow \alpha = 2\theta$$

$$\text{From } \Delta ABC : h = 2l \sin\left(\frac{\alpha}{2}\right) \sin \theta$$

$$h = 2l \sin\left(\frac{\alpha}{2}\right) \sin\left(\frac{\alpha}{2}\right)$$

$$\Rightarrow h = 2l \sin^2\left(\frac{\alpha}{2}\right)$$

Now charge is in equilibrium at point B.

So, using sine rule

$$\Rightarrow \frac{mg}{\sin\left[90 + \frac{\alpha}{2}\right]} = \frac{qE}{\sin[180 - 2\theta]}$$

$$\Rightarrow \frac{mg}{\cos \frac{\alpha}{2}} = \frac{qE}{\sin 2\theta}$$

$$\Rightarrow \frac{mg}{\cos \frac{\alpha}{2}} = \frac{qE}{\sin \alpha} \Rightarrow \frac{mg}{\cos \frac{\alpha}{2}} = \frac{qE}{2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}}$$

$$\Rightarrow qE = mg 2 \sin\left(\frac{\alpha}{2}\right)$$

$$\Rightarrow \frac{q^2 k p}{[2l \sin \frac{\alpha}{2}]^3} = mg 2 \sin\left(\frac{\alpha}{2}\right)$$

$$\Rightarrow \frac{k p q}{[2l \sin \frac{\alpha}{2}]^2} = mg \sin\left(\frac{\alpha}{2}\right) \times \left(2l \sin \frac{\alpha}{2}\right)$$

$$\Rightarrow \frac{k p q}{[2l \sin \frac{\alpha}{2}]^2} = mgh \Rightarrow \text{substituting this in}$$

equation (i)

$$U_f = mgh + \frac{k p q}{[2l \sin \frac{\alpha}{2}]^2}$$

$$\Rightarrow U_f = 2mgh$$

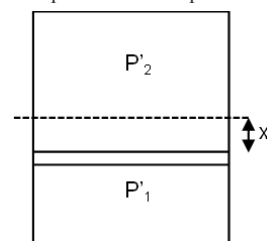
$$W = \Delta U = Nmgh = N = 2$$

6. **Ans (6)**

Assuming temperature remains constant at 300 K

From $P_1 V_1 = P_2 V_2$

$$\frac{P_1 \left(\frac{V_0}{2}\right)}{T} = \frac{P'_1 \left(\frac{V_0}{2} - Ax\right)}{T}$$



$$(P'_1 - P'_2) A = mg$$

$$\left[\frac{P_1 \left(\frac{V_0}{2}\right)}{\frac{V_0}{2} - Ax} - \frac{P_2 \left(\frac{V_0}{2}\right)}{\frac{V_0}{2} + Ax} \right] A = mg$$

$$nRT \left[\frac{1}{4 - x} - \frac{1}{4 + x} \right] = mg$$

$$(0.1)(8.3) \left[\frac{4 + x - 4 + x}{16 - x^2} \right] = mg$$

$$3 \left(\frac{2x}{16 - x^2} \right) = 1$$

$$6x = 16 - x^2$$

$$x^2 + 6x - 16 = 0$$

$$x = 2$$

$$\text{distance} = 4 + 2 = 6 \text{ m}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A,B,C)

Paramagnetic compound (X) are attracted towards magnetic field and the pan is deflected downwards.

While the **Diamagnetic compound (X)** are repelled by magnetic field and pan is deflected upward.

(A) $X \Rightarrow H_2O \rightarrow$ **Diamagnetic** (correct)

(B) $X \Rightarrow K_4[Fe(CN)_6](s) \rightarrow$ **Diamagnetic** (correct)

Here $Fe^{2+} + \text{Strong field ligand} \rightarrow 3d^6 \Rightarrow [t_2g^6, eg^0]$

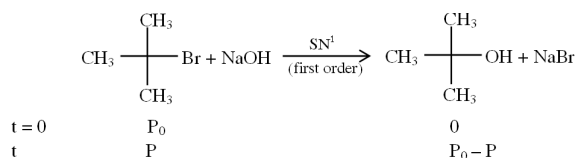
(C) $X \Rightarrow O_2 \rightarrow$ **Paramagnetic** (correct)

Here $O_2(g)$ is paramagnetic due to two-unpaired electrons present in π^* (antibonding orbitals).

(D) $X \Rightarrow C_6H_6(l) \rightarrow$ **Diamagnetic** (Incorrect)

It is due to presence of 0 unpaired electrons.

2. Ans (A)

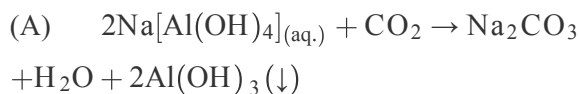


$$\text{rate} = k \left[\text{C(CH}_3)_3\text{Br} \right] \quad \ln \frac{P_0}{P} = kt$$

$$t_{1/2} = \frac{0.693}{k} \quad \ln \frac{P}{P_0} = -kt$$

$$\frac{[Q]}{[P]_0} = \frac{[P_0] - [P]}{[P_0]} = 1 - \frac{[P]}{[P_0]} = 1 - e^{-kt}$$

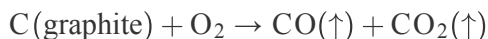
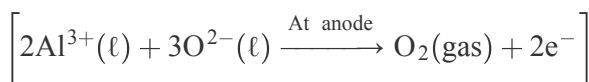
3. Ans (A,B,C,D)



or
 $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}(\text{ppt})$

(B) Function of Na_3AlF_6 is to lower the melting point of electrolyte.

(C) During electrolysis of Al_2O_3 , the reactions at anode are :

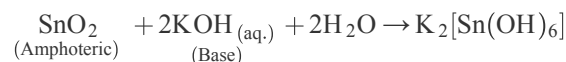


(D) The steel vessel with a lining of carbon acts as cathode.

4. Ans (A,B)

(A) $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ is a reducing agent since Sn^{2+} tends to convert into Sn^{4+} .

(B)



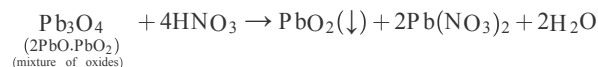
(C) First group cations (Pb^{2+}) form insoluble chloride with HCl that is PbCl_2 however it is slightly soluble in water and therefore lead +2 ion is never completely precipitated on adding hydrochloric acid in test sample of Pb^{2+} , rest of the Pb^{2+} ions are quantitatively precipitated with H_2S in acidic medium.

So that we can say that filtrate of first group contain solution of PbCl_2 in HCl which contains Pb^{2+} and Cl^-

However in the presence of conc. HCl or excess HCl it can produce $\text{H}_2[\text{PbCl}_4]$

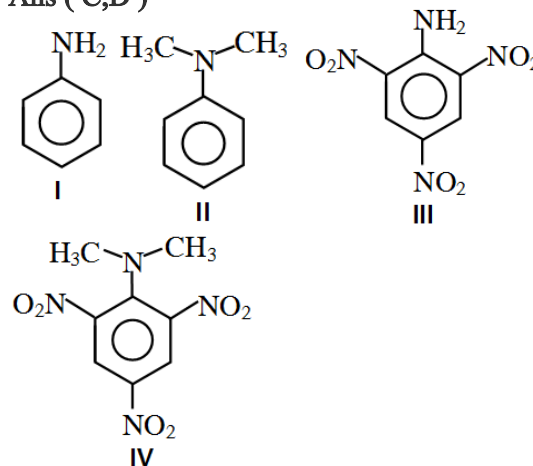
So, we can conclude A, B or A,B,C should be answers.

(D)



It is not a redox reaction.

5. Ans (C,D)



pK_b different between I and II is 0.53 and that of III and IV is 4.6.

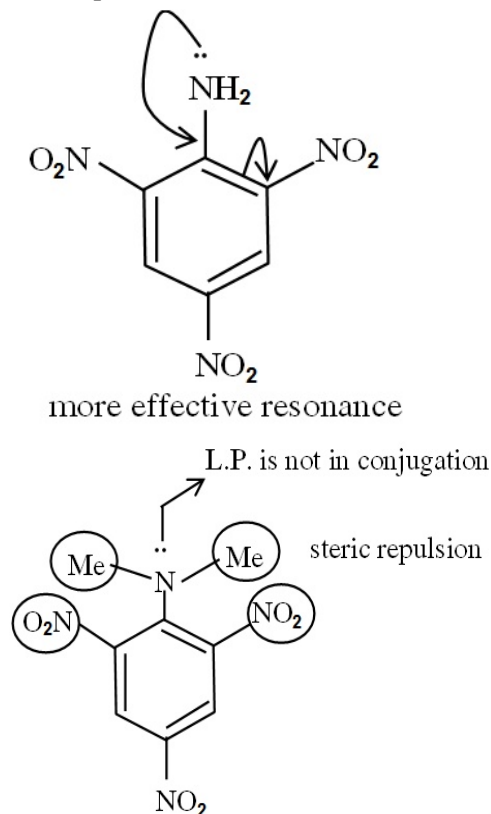
So option (B) is incorrect

Correct Statement (C), (D)

The most basic compound in the given option is (II) and least basic compound is (III)

In 2,4,6-trinitro aniline (III) due to strong $-\text{R}$ effect of $-\text{NO}_2$ groups, the l.p. of $-\text{NH}_2$ is more involved with benzene ring hence it has least basic strength.

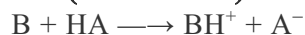
Whereas (IV) N,N-Dimethyl 2,4,6-trinitro aniline, due to steric inhibition to resonance (SIR) effect; the lone pair of nitrogen is not in the plane of benzene, hence make it (l.p.) more free to protonate



PART-2 : CHEMISTRY

SECTION-II

1. Ans (2.80 to 3.20)



$$0.1 \text{ M}, V \text{ ml}$$

$$0.1 \text{ V m mol} \quad 0.1 \text{ V m mol} \quad 0.1 \text{ V} \quad 0.1 \text{ V}$$

$$[BH^+] =$$

$$\frac{0.1 \text{ V}}{2 \text{ V}} = 0.05 \text{ M}$$

$$\text{pH at eq. pt} = 6 \text{ to } 6.28$$

$$\text{pH} = 7 - \frac{1}{2} [\text{p}K_b + \log 0.05]$$

$$\text{So } \text{p}K_b = 2.30 - 2.80$$

Possible

Solution-2

$$\text{at } V = 6 \text{ ml} \quad \text{rxn is complete}$$

$$\text{So } V = 3 \text{ ml} \quad \text{is half of eq. pt}$$

$$\text{at which} \quad \text{pH} = 11$$

$$\text{pOH} = (14 - 11) = \text{p}K_b + \log 1$$

$$\text{p}K_b = 3$$

2. Ans (0.20)

$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$0.3 = P_A^\circ \times 0.25 + P_B^\circ \times 0.75 \dots (i)$$

$$0.4 = P_A^\circ \times 0.5 + P_B^\circ \times 0.5$$

$$0.8 = P_A^\circ + P_B^\circ \dots (ii)$$

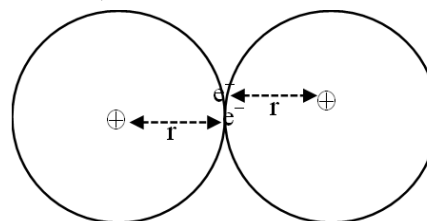
on solving eqⁿ (i) & (ii)

$$P_A^\circ = 0.6, \quad P_B^\circ = 0.2$$

3. Ans (-2640.00--2620.00 or -5280.00--5240.00)

At $d = d_0$, nucleus-nucleus & electron-electron repulsion is absent.

Hence potential energy will be calculated for 2 H atoms. (P.E. due to attraction of proton & electron)



$$P.E. = \frac{-Kq_1q_2}{r} = \frac{(9 \times 10^9)(1.6 \times 10^{-19})^2}{0.529 \times 10^{-10}}$$

(Bohr radius)

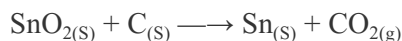
$$= -4.355 \times 10^{-21} \text{ kJ}$$

$$\text{For 1 mol} = -4.355 \times 10^{-21} \times 6.023 \times 10^{23}$$

$$= -2623.249 \text{ kJ/mol}$$

$$\text{For 2 H atoms} = -5246.49 \text{ kJ/mol}$$

5. **Ans (935.00)**



$$\Delta H^\circ_{\text{rxn}} = [-394] - [-581] = 187 \text{ kJ/mole}$$

$$\Delta S^\circ_{\text{rxn}} = [52 + 210] - [56 + 6] = 200 \text{ J/k-mole}$$

$$T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{187 \times 1000}{200} = 935\text{K}$$

6. **Ans (0.20)**

$$\begin{aligned} \text{For ppt, } [\text{Zn}^{+2}][\text{S}^{-2}] &= K_{\text{sp}} \frac{1.25 \times 10^{-22}}{0.05} \\ [\text{S}^{-2}] &= \frac{2.5 \times 10^{-21} \text{MH}_2\text{S} \rightleftharpoons 2\text{H}^+ + \text{S}^{-2} K_{\text{Net}} = 10^{-21}}{0.1} \\ &= \frac{[\text{H}^+]^2 \times 2.5 \times 10^{-21}}{0.1} \\ [\text{H}^+]^2 &= \frac{1}{25} \\ [\text{H}^+] &= \frac{1}{5} \text{M} = 0.2 \text{M} \end{aligned}$$

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (9)**

Atomic Number	Ionization Enthalpy (kJ/mol)		
	I ₁	I ₂	I ₃
n	1681	3374	6050
n + 1	2081	3952	6122
n + 2	496	4562	6910
n + 3	738	1451	7733

By observing the values of I₁, I₂ & I₃ for atomic number (n+2), it is observed that I₂ >> I₁.

This indicates that number of valence shell electrons is 1 and atomic number (n+2) should be an alkali metal.

Also for atomic number (n+3), I₃ >> I₂.

This indicates that it will be an alkaline earth metal which suggests that atomic number (n+1) should be a noble gas & atomic number (n) should belong to Halogen family. Since n < 10; hence

$$n = 9 \text{ (F atom)}$$

Note : n = 1 (H atom) cannot be the answer because it does not have I₂ & I₃ values.

2. **Ans (6)**

Here polar molecules in the liquid form will be attracted/deflected near charged comb.

Polar molecules : HF, H₂O, NH₃, H₂O₂, CHCl₃, C₆H₅Cl (6-polar molecules)

Nonpolar molecules : O₂, CCl₄, C₆H₆

3. **Ans (6)**

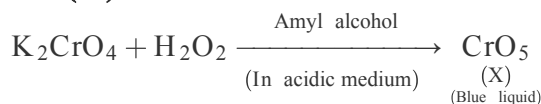


$$\text{Eq of KMnO}_4 = \text{Eq of I}_2$$

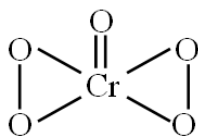
$$4 \times 3 = n \times 2$$

$$n = 6$$

4. Ans (4)



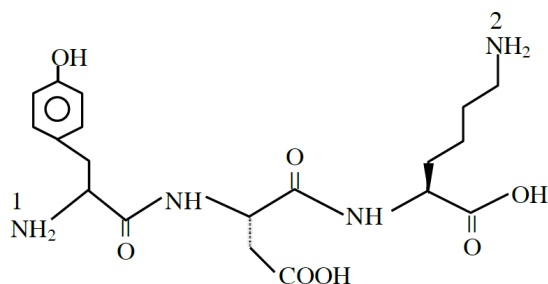
Here the structure of CrO_5 is :-



Here, single bonded O-atoms with Cr is = 04

5. Ans (5)

$$|z_1| + |z_2| + |z_3| = 5$$



At pH = 2 $\overset{1}{\text{N}}\text{H}_2$ and $\overset{2}{\text{N}}\text{H}_2$ of Tyrosine and Lysine is +ve charged (+1 each)

$$+2 |z_1| = 2$$

At pH = 6 NH_2 of Lysine (+1),

COOH (-1) of glutamic acid, so because of dipolar ion exist $|z_2| = 0$

At pH = 11 COOH of Glutamic acid (-1)

COOH of Lysine (-1)

OH of phenol (-1)

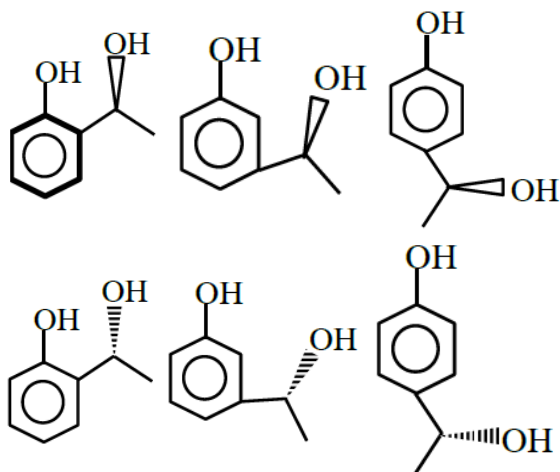
$$|z_3| = 3$$

6. Ans (6)

$\text{C}_8\text{H}_{10}\text{O}_2 \rightarrow$ Gives FeCl_3 test means Phenol derivative



Rotate plane polarized light means optically active



PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,C)

$$f'(x) = \frac{f(x)}{b^2 + x^2}$$

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{dx}{x^2 + b^2}$$

$$\Rightarrow \ln |f(x)| = \frac{1}{b} \tan^{-1} \left(\frac{x}{b} \right) + c$$

$$\text{Now } f(0) = 1$$

$$\therefore c = 0$$

$$\therefore |f(x)| = e^{\frac{1}{b} \tan^{-1} \left(\frac{x}{b} \right)}$$

$$\Rightarrow f(x) = \pm e^{\frac{1}{b} \tan^{-1} \left(\frac{x}{b} \right)}$$

$$\text{since } f(0) = 1 \therefore f(x) = e^{\frac{1}{b} \tan^{-1} \left(\frac{x}{b} \right)}$$

$$x \rightarrow -x$$

$$f(-x) = e^{-\frac{1}{b} \tan^{-1} \left(\frac{x}{b} \right)}$$

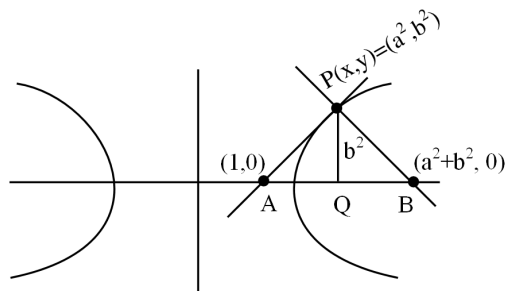
$$\therefore f(x) \cdot f(-x) = e^0 = 1 \text{ (option C)}$$

and for $b > 0$

$$f(x) = e^{\frac{1}{b} \tan^{-1} \left(\frac{x}{b} \right)}$$

$\Rightarrow f(x)$ is increasing for all $x \in \mathbb{R}$ (option A)

2. Ans (A,D)



Since Normal at point P makes equal intercept on co-ordinate axes, therefore slope of Normal = -1

Hence slope of tangent = 1

Equation of tangent

$$y - 0 = 1(x - 1)$$

$$y = x - 1$$

Equation of tangent at (x_1, y_1)

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

$$x - y = 1 \text{ (equation of Tangent)}$$

on comparing $x_1 = a^2, y_1 = b^2$

$$\text{Also } a^2 - b^2 = 1 \quad \dots(1)$$

Now equation of normal at $(x_1, y_1) = (a^2, b^2)$

$$y - b^2 = -1(x - a^2)$$

$$x + y = a^2 + b^2 \quad \dots(\text{Normal})$$

point of intersection with x-axis is $(a^2 + b^2)$

$$\text{Now } e = \sqrt{1 + \frac{b^2}{a^2}}$$

$$e = \sqrt{1 + \frac{b^2}{b^2 + 1}} \quad \left[\text{from (1)} \quad \frac{b^2}{b^2 + 1} < 1 \right]$$

$$1 < e < \sqrt{2} \quad \text{option (A)}$$

$$\Delta = \frac{1}{2} \cdot AB \cdot PQ$$

$$\text{and } \Delta = \frac{1}{2} (a^2 + b^2 - 1) \cdot b^2$$

$$\Delta = \frac{1}{2} (2b^2) b^2 \quad (\text{from (1)} \quad a^2 - 1 = b^2)$$

$$\Delta = b^4 \text{ so option (D)}$$

3. Ans (A,B,D)

since $f(x) = xg(x)$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} xg(x)$$

$$\lim_{x \rightarrow 0} f(x) = \left(\lim_{x \rightarrow 0} x \right) \cdot \left(\lim_{x \rightarrow 0} g(x) \right)$$

$$\lim_{x \rightarrow 0} f(x) = 0 \times 1 = 0 \quad \dots(1)$$

$$f(x+y) = f(x) + f(y) + f(x)f(y)$$

Now we check continuity of $f(x)$

at $x = a$

$$\lim_{h \rightarrow 0} f(a+h) = f(a) + f(h) + f(a)f(h)$$

$$\lim_{x \rightarrow 0} (f(a) + f(h) (1 + f(a)))$$

$$\lim_{h \rightarrow 0} f(a+h) = f(a)$$

$\therefore f(x)$ is continuous $\forall x \in \mathbb{R}$

$$\lim_{x \rightarrow 0} f(x) = f(0) = 0 \quad \left(\lim_{x \rightarrow 0} f(x) = 0 \right)$$

$$\therefore f(0) = 0$$

$$\text{and } \lim_{x \rightarrow 0} \frac{f'(x)}{1} = 1$$

$$\therefore f'(0) = 1$$

Now

$$f(x+y) = f(x) + f(y) + f(x)f(y)$$

using partial derivative (w.r.t. y)

$$f'(x+y) + f'(y) + f(x) + f'(y)$$

put $y = 0$

$$f'(x) = f'(0) + f(x) f'(0)$$

$$f'(x) = 1 + f(x)$$

$$\int \frac{f'(x)}{1 + f(x)} dx = \int 1 dx$$

$$\ln |1 + f(x)| = x + C$$

$$f(0) = 0; c = 0 \therefore |1 + f(x)| = e^x$$

$$1 + f(x) = \pm e^x \text{ or } f(x) = \pm e^x - 1$$

$$\text{Now } f(0) = 0 \therefore f(x) = e^x - 1$$

$$\therefore f(x) = e^x - 1$$

option (A) is correct

$$\text{and } f'(x) = e^x$$

$$f'(0) = 1 \text{ option (D) is correct}$$

$$g(x) = \frac{f(x)}{x} = \begin{cases} \frac{e^x - 1}{x} & ; x \neq 0 \\ 1 & ; x = 0 \end{cases}$$

$$g'(0+h) = \lim_{h \rightarrow 0} \frac{g(0+h) - g(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{e^h - 1}{h} - 1}{h} = \frac{1}{2}$$

option B is correct

4. Ans (A,B,C)

R is mid point of PQ

$$\therefore R(2, 1, -1)$$

and it lies on

plane

equation of

plane is $\alpha x + \beta y$

$$+ \gamma z = \delta$$

$$\therefore 2\alpha + \beta - \gamma = \delta$$

$$\dots(1)$$

Normal vector to plane is

$$\vec{n} = 2\vec{i} + 2\vec{j}$$

$$\therefore \frac{\alpha}{2} = \frac{\beta}{2} = \frac{\gamma}{0} = k$$

$$\therefore \alpha = 2k, \beta = 2k, \gamma = 0 \dots(2)$$

$$\text{and } \alpha + \gamma = 1 \text{ (given)} \dots(3)$$

from (2) and (3)

$$\therefore \alpha = 1, \beta = 1, \gamma = 0$$

and from (1)

$$2(1) + 1 - 0 = \delta$$

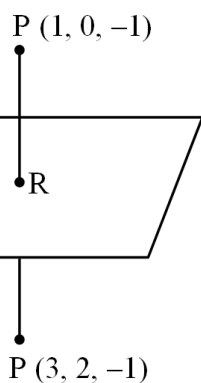
$$\delta = 3$$

Now :

$$\alpha + \beta = 2$$

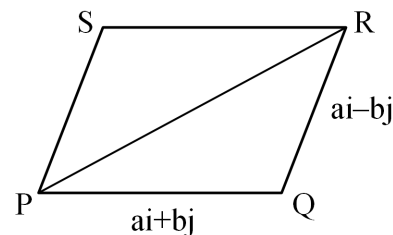
$$\delta - \gamma = 3$$

$$\delta + \beta = 4$$



5. Ans (A,C)

$$\vec{u} = ((\vec{i} + \vec{j}) \cdot \widehat{PQ}) \widehat{PQ}$$



$$\vec{u} = |(\vec{i} + \vec{j}) \cdot \widehat{PQ}|$$

$$|\vec{u}| = \left| (\vec{i} + \vec{j}) \cdot \frac{(a\vec{i} + b\vec{j})}{\sqrt{a^2 + b^2}} \right| = \frac{a + b}{\sqrt{a^2 + b^2}}$$

$$\vec{v} = (\vec{i} + \vec{j}) \cdot \widehat{PS}$$

$$|\vec{v}| = \left| (\vec{i} + \vec{j}) \cdot \frac{(a\vec{i} - b\vec{j})}{\sqrt{a^2 + b^2}} \right| = \frac{a - b}{\sqrt{a^2 + b^2}}$$

$$|\vec{u}| = |\vec{v}| = |\vec{w}|$$

$$\frac{|(a + b)| + |(a - b)|}{\sqrt{a^2 + b^2}} = \sqrt{2}$$

For $a > b$

$$2a = \sqrt{2} \cdot \sqrt{a^2 + b^2}$$

$$4a^2 = 2a^2 + 2b^2$$

$$a^2 = b^2 \therefore a = b \dots(1)$$

$$(a > 0, b > 0)$$

similarly for $a < b$ we will get $a = b$

Now area of parallelogram

$$= |(a\vec{i} + b\vec{j}) \times (a\vec{i} - b\vec{j})|$$

$$= 2ab$$

$$\therefore 2ab = 8$$

$$ab = 4 \dots(2)$$

from (1) and (2)

$$a = 2, b = 2 \therefore a + b = 4 \text{ option (A)}$$

length of diagonal is

$$|2a\hat{i}| = |4\hat{i}| = 4$$

6. Ans (A,B,D)

Solving

$$f(m, n, p) = \sum_{i=0}^p {}^m C_i \cdot {}^{n+i} C_p \cdot {}^{p+n} C_{p-i}$$

$${}^m C_i \cdot {}^{n+i} C_p \cdot {}^{p+n} C_{p-i}$$

$${}^m C_i \cdot \frac{(n+i)!}{p!(n-p+i)!} \times \frac{(n+p)!}{(p-i)!(n+i)!}$$

$${}^m C_i \times \frac{(n+p)!}{p!} \times \frac{1}{(n-p+i)!(p-i)!}$$

$${}^m C_i \times \frac{(n+p)!}{p!n!} \times \frac{n!}{(n-p+i)!(p-i)!}$$

$${}^m C_i \cdot {}^{n+p} C_p \cdot {}^n C_{p-i} \quad \{ {}^m C_i \cdot {}^n C_{p-i} = {}^{m+n} C_p \}$$

$$f(m, n, p) = {}^{n+p} C_p \cdot {}^{m+n} C_p$$

$$\frac{f(m, n, p)}{{}^{n+p} C_p} = {}^{m+n} C_p$$

Now

$$g(m, n) = \sum_{p=0}^{m+n} \frac{f(m, n, p)}{{}^{n+p} C_p}$$

$$g(m, n) = \sum_{p=0}^{m+n} {}^{m+n} C_p$$

$$g(m, n) = 2^{m+n}$$

$$(A) g(m, n) = q(n, m)$$

$$(B) g(m, n+1) = 2^{m+n+1}$$

$$g(m+n, n) = 2^{m+1+n}$$

$$(D) g(2m, 2n) = 2^{2m+2n}$$

$$= (2^{m+n})^2$$

$$= (g(m, n))^2$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (495.00)

Selection of 4 days out of 15 days such that no two of them are consecutive

$$= {}^{15-4+1} C_4 = {}^{12} C_4$$

$$= \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2} = 11 \times 5 \times 9 = 495$$

2. Ans (1080.00)

$$\text{required ways} = \frac{6!}{2! 2! 1! 1! 2! 2!} \times 4! = 1080$$

3. Ans (8.00)

Prime : 2, 3, 5, 7, 11

$$1 \quad 2 \quad 4 \quad 6 \quad 2$$

$$P(\text{Prime}) = \frac{15}{36}$$

Perfect square = 4, 9

$$3 \quad 4$$

$$P(\text{perfect square}) = \frac{7}{36}$$

required probability

$$= \frac{\frac{4}{36} + \frac{14}{36} \times \frac{4}{36} + \left(\frac{14}{36}\right)^2 \frac{4}{36} + \dots}{\frac{7}{36} + \frac{14}{36} \times \frac{7}{36} + \left(\frac{14}{36}\right)^2 \frac{7}{36} + \dots}$$

$$P = \frac{4}{7}$$

$$\therefore 14P = 14 \cdot \frac{4}{7} = 8$$

4. Ans (19.00)

$$f(x) + f(1-x) = \frac{4^x}{4^x + 2} + \frac{4^{1-x}}{4^{1-x} + 2}$$

$$= \frac{4^x}{4^x + 2} + \frac{4/4^x}{\frac{4}{4^x} + 2}$$

$$= \frac{4^x}{4^x + 2} + \frac{4}{4 + 2 \cdot 4^x}$$

$$= \frac{\frac{4^x}{4^x}}{\frac{4^x}{4^x} + 2} + \frac{2}{2 + 4^x}$$

so,

$$f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + \dots + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$$

$$= 19 + f\left(\frac{1}{2}\right) - f\left(\frac{1}{2}\right) = 19$$

5. **Ans (4.00)**

$$F(x) = \int_0^x f(t) \cdot dt$$

$$\Rightarrow F'(x) = f(x)$$

$$I = \int_0^{\pi} f'(x) \cdot \cos x \, dx + \int_0^{\pi} F(x) \cos(x) \, dx = 2 \quad \dots(1)$$

$$I_1 = \int_0^{\pi} f'(x) \cdot \cos x \, dx \text{ (Let)}$$

Using by parts

$$I_1 = (\cos x \cdot f(x)) \Big|_0^{\pi} + \int_0^{\pi} \sin x \cdot f(x) \, dx$$

$$I_1 = 6 - f(0) + \int_0^{\pi} \sin x \cdot F'(x) \, dx$$

$$I_1 = 6 - f(0) + I_2 \quad \dots(2)$$

$$I_2 = \int_0^{\pi} \sin x \cdot F'(x) \cdot dx$$

Using by part we get

$$I_2 = (\sin x \cdot F(x)) \Big|_0^{\pi} - \int_0^{\pi} \cos x \cdot F(x) \, dx$$

$$I_2 = - \int_0^{\pi} \cos x \cdot F(x) \, dx$$

$$(2) \Rightarrow I_1 = 6 - f(0) - \int_0^{\pi} \cos x \cdot F(x) \, dx$$

$$(1) \Rightarrow I = 6 - f(0) = 2 \Rightarrow f(0) = 4$$

6. **Ans (0.50)**

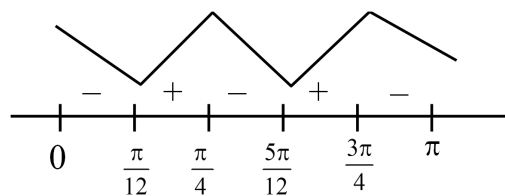
$$f(\theta) = (\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^4$$

$$f(\theta) = \sin^2 2\theta - \sin 2\theta + 2$$

$$f'(\theta) = 2(\sin 2\theta) \cdot (2\cos 2\theta) - 2\cos 2\theta$$

$$= 2\cos 2\theta(2\sin 2\theta - 1)$$

critical points



$$\text{so, minimum at } \theta = \frac{\pi}{12}, \frac{5\pi}{12}$$

$$\lambda_1 + \lambda_2 = \frac{1}{12} + \frac{5}{12} = \frac{6}{12} = \frac{1}{2}$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (8)**

$$\text{Let } z = x + iy$$

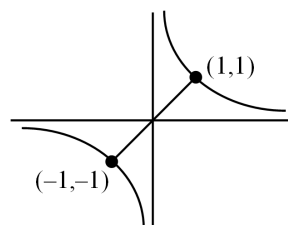
$$z^4 - |z|^4 = 4iz^2$$

$$\Rightarrow z^4 - (z\bar{z})^2 = 4iz^2$$

$$\Rightarrow z = 0 \text{ or } z^2 - (\bar{z})^2 = 4i$$

$$\Rightarrow 4ixy = 4i$$

$$\Rightarrow xy = 1$$



$$|z_1 - z_2|_{\min}^2 = 8$$

2. **Ans (6)**

Let $P(r)$ = probability of r successes

$$= {}^nC_r \left(\frac{3}{4}\right)^r \left(\frac{1}{4}\right)^{n-r}$$

$$1 - (P(0) + P(1) + P(2)) \geq 0.95$$

$$\Rightarrow 1 - {}^nC_0 \left(\frac{1}{4}\right)^n - {}^nC_1 \left(\frac{3}{4}\right) \left(\frac{1}{4}\right)^{n-1} - {}^nC_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^{n-2} \geq 0.95$$

$$\Rightarrow 1 - \left(\frac{1 + 3n + \frac{9n(n-1)}{2}}{4^n} \right) \geq 0.95$$

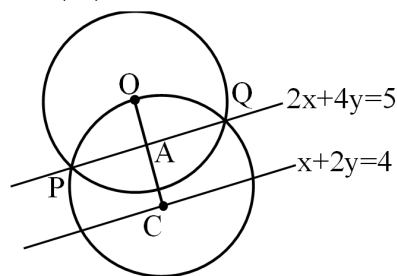
$$\Rightarrow 9n^2 - 3n + 2 \leq 0.05 \times 4^n \times 2 \leq \frac{4^n}{10}$$

$$\text{for } n = 5 \quad 212 \leq 102.4 \quad (\text{Not true})$$

$$\text{for } n = 6 \quad 308 \leq 409.6 \quad \text{true}$$

$$\therefore \text{least value of } n = 6$$

3. Ans (2)



M-I

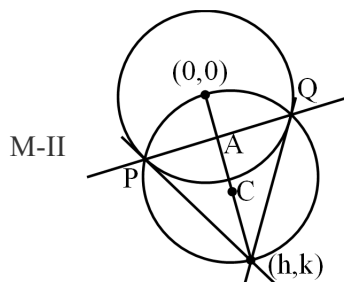
$$OA = \frac{\sqrt{5}}{2} \quad OC = \frac{4}{\sqrt{5}}$$

$$CQ = OC = \frac{4}{\sqrt{5}} \text{ and } CA = \frac{3}{2\sqrt{5}}$$

$$\therefore OQ = \sqrt{OA^2 + AQ^2} = \sqrt{OA^2 + (CQ^2 - CA^2)}$$

$$\Rightarrow \sqrt{\frac{5}{4} + \frac{16}{5} - \frac{9}{20}} = \sqrt{4}$$

$$\Rightarrow 2 = r$$



$$PQ : hx + ky = r^2$$

$$\text{Given } PQ : 2x + 4y = 5$$

$$\Rightarrow \frac{h}{2} = \frac{k}{4} = \frac{r^2}{5} \Rightarrow h = \frac{2r^2}{5} \quad k = \frac{4r^2}{5}$$

$$\therefore C = \left(\frac{r^2}{5}, \frac{2r^2}{5} \right)$$

$$\therefore C \text{ lies on } x + 2y = 4$$

$$\Rightarrow \frac{r^2}{5} + 2 \left(\frac{2r^2}{5} \right) = 4$$

$$\Rightarrow r^2 = 4 \Rightarrow r = 2$$

4. Ans (5)

M-I

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^2 = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + dc & bc + d^2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a^3 + 2abc + bdc & a^2b + abd + b^2c + bd^2 \\ a^2c + adc + bc^2 + d^2c & abc + 2bcd + d^3 \end{bmatrix}$$

$$\text{Given trace}(A) = a + d = 3$$

$$\text{and trace}(A^3) = a^3 + d^3 + 3abc + 3bcd = -18$$

$$\Rightarrow a^3 + d^3 + 3bc(a + d) = -18$$

$$\Rightarrow a^3 + d^3 + 9bc = -18$$

$$\Rightarrow (a + d)((a + d)^2 - 3ad) + 9bc = -18$$

$$\Rightarrow 3(9 - 3ad) + 9bc = -18$$

$$\Rightarrow ad - bc = 5 = \text{determinant of } A$$

M-II

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} ; \Delta = ad - bc$$

$$|A - \lambda I| = (a - \lambda)(d - \lambda) - bc$$

$$= \lambda^2 - (a + d)\lambda + ad - bc$$

$$= \lambda^2 - 3\lambda + \Delta$$

$$\Rightarrow O = A^2 - 3A + \Delta I$$

$$\Rightarrow A^2 = 3A - \Delta I$$

$$\Rightarrow A^3 = 3A^2 - \Delta A$$

$$= 3(3A - \Delta I) - \Delta A$$

$$= (9 - \Delta)A - 3\Delta I$$

$$= (9 - \Delta) \begin{bmatrix} a & b \\ c & d \end{bmatrix} - 3\Delta \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore \text{trace } A^3 = (9 - \Delta)(a + d) - 6\Delta$$

$$\Rightarrow -18 = (9 - \Delta)(3) - 6\Delta$$

$$= 27 - 9\Delta$$

$$\Rightarrow 9\Delta = 45 \Rightarrow \Delta = 5$$

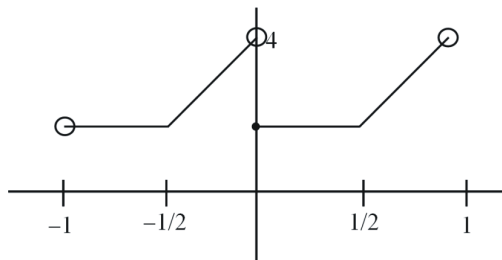
5. Ans (4)

$$f(x) = |2x - 1| + |2x + 1|$$

$$g(x) = \{x\}$$

$$f(g(x)) = |2\{x\} - 1| + |2\{x\} + 1|$$

$$= \begin{cases} 2 & \{x\} \leq \frac{1}{2} \\ 4\{x\} & \{x\} > \frac{1}{2} \end{cases}$$



discontinuous at $x = 0 \Rightarrow c = 1$

Non differential at $x = -\frac{1}{2}, 0, \frac{1}{2} \Rightarrow d = 3$

$$\therefore c + d = 4$$

6. Ans (8)

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2} \cdot 2 \sin 2x \cos x}{2 \sin 2x \sin \frac{3x}{2} + (\cos \frac{5x}{2} - \cos \frac{3x}{2}) - \sqrt{2}(1 + \cos 2x)}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin x \cos^2 x}{2 \sin 2x (\sin \frac{3x}{2} - \sin \frac{x}{2}) - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin x \cos^2 x}{4 \sin x \cos x (2 \cos x \cdot \sin \frac{x}{2}) - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin x}{8 \sin x \cdot \sin \frac{x}{2} - 2\sqrt{2}} = 8$$